DATA Classification - who guideline in coding

health records and information technology (Kenya Medical Training College)

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DATA CLAASIFICATION

1. Define disease classification (icd) & (icpm)
2. Historical development of icd
3. Explain type of classification of disease operation & procedure
4. Explain purpose of disease classification operation & procedure
5. List & describe the 22 major type of icd
6. Describe the arrangement of icd & icpm
7. Explain the procedure of indexing & coding
8. Describe abbreviation, convention, punctuation as used in icd & icpm
9. Define & describe the diagnostic indexing
10. Demonstrate ability code & indexing & medical procedure in icd & icpm
11. Demonstrate ability to create manual and electric indexing
12. Describe various medical term used coding & classification of disease
13. Differentiate between primary & secondary diagnostic & dual classification
14. Display tiredness, legibility & accuracy in completing indexing
15. Demonstrate ability to edify audit case record
16. Demonstrate ability to generate analyze & interpreter the information
17. Describe the procedure in data security, confidentiality & ethics (data protection)
18. Certification
19. Rules an d guidelines for morbidity and mortality
20. Statistical presentation

Definition of disease classification (icd) & (icpm)

- ***A classification of diseases*** *can be defined as a system of categories to which morbid entities are assigned according to established criteria.*
- ***Classification*** *is the use of police procedure to categories fact which stored for accessible use.*

Purpose of icd;

- Is to *permit the systematic recording analysis, interpretation and comparison of mortality and morbidity data collected in different countries or areas and at different times.*
- *Comparison national morbidity and mortality data collection at different time.*

THE ICD

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- *Icd is the international standard diagnostic classification for all general epidemiological and many health management purposes*.
- These include the analysis of the general health situation of population groups and the monitoring of the incidence and prevalence of diseases and other health problems in relation to other variables, such as the characteristics and circumstances of the individuals affected.

The ICD can be used to

- ✓ ***Classify diseases*** and other health problems recorded on many types of health and vital records
- ✓ To ***translate diagnoses*** *of diseases and other health problems from words into an alphanumeric code, this permits easy storage, retrieval and analysis of the data.*
- ✓ Provide designing healthy delivery system.
- ✓ Setting healthy care policy through the use of precise input (treatment) & output (outcome) data.
- ✓ Monitor pattern & levels of utilization for designing payment system.
- ✓ Organization of information. evaluation of new treatment drugs
- ✓ To classify causes of mortality as recorded at the registration of death & diagnoses in morbidity.
- ✓ Provides for a wide variety of signs, symptoms, abnormal findings, complaints, and social circumstances that may stand in place of a diagnosis on health-related records
- ✓ used to classify data recorded under headings such as "diagnosis", "reason for admission", "conditions treated" and "reason for consultation", which appear on a wide variety of health records from which statistics and other health-situation information are derived. [1]

Objectives of data classification

- Equip a student with knowledge, skills & attitude that will enable him/ her to code & index disease, conditions & procedure in medicine necessary for production of standard morbidity & mortality information.

Purposes of disease classification

- ✓ Help in analyzing general health situation-inference-generalizing
- ✓ Help in clinical & research purpose-monitoring
- ✓ Providing a mean & classifying medical terminology
- ✓ Categorize disease for morbidity & mortality reporting
- ✓ It promote international comparability in the collation & classification of disease

Uses of medical data classification

- ✓ Help in standardizing health uniformity worldwide

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- ✓ Provide a base for research activity interest
- ✓ Provider of a tool for teaching purpose for all health profession
- ✓ Provide a tool for health service management on daily geographical
- ✓ Provides a base for planning health service on short & long term
- ✓ Classification is a method of generalization.

The basic structure and principles of classification of the ICD

(ACODING TO FARR) in Paris in 1855

The structure has developed out of that proposed by William Farr in the early days of international discussions on classification structure. His scheme was that, for all practical, epidemiological purposes, statistical data on diseases should be grouped in the following way:

- ✓ *Epidemic diseases,*
- ✓ *Constitutional (general) diseases,*
- ✓ *Local diseases arranged according to anatomical site,*
- ✓ *Developmental diseases*
- ✓ *Diseases that are the direct result of violence / injury*

Type of disease classification

- ✓ *Topography*
- ✓ *Statistic*
- ✓ *Anatomic by organ or tissue*
- ✓ *Physiological by function or effect*
- ✓ *Pathological by nature of disease*

Principle of identifying disease classification

- ✓ ***Malignant*** –single condition/ general, harmful & tend to produce death.
- ✓ ***Situ***-disease that does not spread
- ✓ ***Benign***- not posing serious treat
- ✓ ***Metastatic***-can spread faster to other parts of the body.
- ✓ ***Primary site***-initial part of the body affected
- ✓ ***Secondary site***-part of the body where a disease has been transferred to.

Roles/ responsibilities of WHO in implementing ICD 10

- ✓ WHO help in ***Provides guidance*** and ***advice***

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- ✓ Promotes the ***development of adaptations*** that extend both the usefulness of the ICD
- ✓ ***WHO help in Comparability*** of health statistics.
- ✓ Development of new classifications, adaptations, and glossaries is to ***provide cooperative leadership***
- ✓ Act as a ***clearing-house***, giving technical advice, and guidance and support when needed.

HISTORICAL DEVELOPMENT OF ICD

PERSONS	PERIOD	TITLE OF WORK	CLASSICATION
Sir George Knibbs	***;;;;;;;;;;;;;;;;;;;***	***;;;;;;;;;;;;;;;;;;;;;***	***;;;;;;;;;;;;;;;;;;;

- Of ***Edinburgh,***
- His work was published in ***1785*** under the title ***Synopsis nosologiae methodicae***.

3. ***<u>John Graunt</u>***

- For all ***practical purposes***
- The statistical study of disease began a century earlier with the work of ***John Graunt*** on *the London Bills of Mortality.*
- The kind of classification envisaged by this pioneer is exemplified by his *attempt to estimate the proportion of live born children who died before reaching the age of six years, no records of age at death being available. He took all deaths classed as thrush, convulsions, rickets, teeth and worms, abortive, chrisoms, infants, liver grown, and overlaid and added to them half the deaths classed as smallpox, swinepox, measles, and worms without convulsions. Despite the crudity of this classification his estimate of 36% mortality before the age of six years appears from later evidence to have been a good one*.

4. ***<u>William Farr</u>***

- While three centuries have contributed something to the scientific accuracy of disease classification, there are many who doubt the usefulness of attempts to compile statistics of disease, or even causes of death, because of the difficulties of classification. To these, one can quote Major Greenwood: "The scientific purist, who will wait for medical statistics until they are nosologically exact, is no wiser than Horace's rustic waiting for the river to flow away
- Fortunately for the progress of preventive medicine, the General Register Office of England and Wales, at its inception in 1837, found in

<u>William Farr (1807-1883) –</u>

- ***First medical statistician***
- ***Initiative that William farr is well known for'***

He made the best possible use of the imperfect classifications of disease available at the time

Labored to secure better classifications and international uniformity in their use

Wrote principles that should govern

- Farr found the *classification of Cullen in use in the public services of his* day. It had *not been revised to embody the advances of medical science,* nor was it deemed by him to be satisfactory for statistical purposes.
- In the first Annual Report of the Registrar General, *therefore, he discussed the principles that should govern a statistical classification of disease and urged the adoption of a uniform classification as follows:*

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- **The advantages of a uniform statistical nomenclature, however imperfect, are so obvious, that it is surprising no attention has been paid to its enforcement in Bills of Mortality. Each disease has, in many instances, been denoted by three or four terms, and each term has been applied to as many different diseases: vague, inconvenient names have been employed, or complications have been registered instead of primary diseases. The nomenclature is of as much importance in this department of inquiry as weights and measures in the physical sciences, and should be settled without delay.**

- *Both nomenclature and statistical classification received constant study and consideration by Farr in his annual* "Letters" to the Registrar General published in the Annual Reports of the Registrar General.
- The *utility of a uniform classification of causes of death was so strongly recognized at the first International Statistical Congress, held in* ***Brussels in 1853,*** that the Congress requested William Farr and Marc d'Espine, of Geneva, to prepare an internationally applicable, uniform classification of causes of death.
- *At the next Congress*, in ***Paris in 1855***, Farr and d'Espine submitted two separate lists which were based on very different principles.
- Farr's classification was arranged under five groups:
  - ✓ Epidemic diseases,
  - ✓ Constitutional (general) diseases,
  - ✓ Local diseases arranged according to anatomical site,
  - ✓ Developmental diseases
  - ✓ Diseases that are the direct result of violence.
- **D'Espine** classified diseases according to ***their nature*** (*gouty, herpetic, haematic, etc.)*.
- The *Congress adopted a compromise* ***list of 139 rubrics***.
- ***In 1864***, this classification was revised in Paris on the basis of Farr's model and was subsequently further revised in ***1874, 1880, and 1886***. (Formulae 9, 10, 6, 6,)
- Although this classification was never universally accepted, the general arrangement proposed by Farr, including the principle of classifying diseases by anatomical site, survived as the basis of the International List of Causes of Death.

ADOPTION OF THE INTERNATIONAL LIST OF CAUSES OF DEATH

Jacques Bertillon (1851-1922

- ***The International Statistical Institute, the successor to the International Statistical Congress***, at its meeting in ***Vienna in 1891***, charged a committee, chaired by J**acques Bertillon *(1851-1922),*** *Chief of Statistical Services* of the **City of Paris**, *with the preparation of a classification of causes of death*.

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- It is of interest to note that Bertillon was the grandson of ***Achille Guillard***, ***a noted botanist and statistician, who had introduced the resolution requesting Farr and d'Espine to prepare a uniform classification at the first International Statistical Congress in 1853***.
- *The report of this committee was presented by Bertillon at the meeting of the International Statistical Institute in* ***Chicago in 1893*** *and adopted by it*.
- The classification prepared by Bertillon's committee was based on the *classification of causes of death used by the* ***City of Paris***, *which, since its revision in 1885*, represented a synthesis of English, German, and Swiss classifications.
- The classification *was based on the principle, adopted by* Farr,
  - ✓ ***of distinguishing between general diseases***
  - ✓ ***And those localized to a particular organ or anatomical site***.
- In accordance with the instructions of the Vienna Congress made at the suggestion of L. Guillaume, the Director of the Federal Bureau of Statistics of Switzerland, Bertillon ***included three classifications, group line classification or the Bertillon classification causes of death***
  - ✓ An abridged classification of 44 titles
  - ✓ A classification of 99 titles
  - ✓ A classification of 161 titles.
- The Bertillon Classification of Causes of Death, as it was first called, received general approval and was adopted by ***several countries***, as well as by ***many cities***. The classification was first used in ***North America by Jesus E. Monjaras*** for the statistics ***of San Luis de Potosi, Mexico***
- In 1898, ***the American Public Health Association (APHA)***, at its meeting in ***Ottawa***, ***Canada***, recommended the adoption of the Bertillon Classification by registrars of ***Canada, Mexico, and the United States of America***.
- The Association further suggested that the classification should be ***revised every ten years.***
- At the meeting of the International Statistical Institute at ***Christiania in 1899,*** *Bertillon presented a report on the progress of the classification, including the recommendations of the American Public Health Association for decennial revisions.*

The International Statistical Institute then adopted the following resolution/ recommendations;

- *The International Statistical Institute convinced of the necessity of using in the different countries comparable nomenclatures:*
  - ✓ ***Learns with pleasure of the adoption by all the statistical offices*** of North America, by some of those of South America, and by some in Europe, of the system of cause of death nomenclature presented in 1893;
  - ✓ ***Insists vigorously that this system of nomenclature be adopted in principle and without revision***, by all the statistical institutions of Europe;
  - ✓ ***Approves, at least in its general lines****, the system of decennial revision proposed by the American Public Health Association at its Ottawa session (1898):*

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- ✓ ***Urges the statistical offices that have not yet adhered, to do so without delay,*** and *to contribute to the comparability of the cause of death nomenclature.*

- The French Government therefore convoked in ***Paris, in August 1900***, *the first International Conference for the Revision of the Bertillon or International List of Causes of Death*.
- ***Delegates from 26 countries*** attended this Conference.
- *A detailed classification of causes of death consisting of* ***179 groups*** *and an abridged classification of 35 groups were adopted on 21 August 1900.*
- The desirability of decennial revisions was recognized, *and the French Government was requested to call the next meeting in 1910*.
- In fact the next conference was held in ***1909***, and the *Government of France* called succeeding conferences in ***1920, 1929, and 1938. (***FOMULAE 9, 11, 9, 9)
- Bertillon continued to be the guiding force in the promotion of the International List of Causes of Death, and the revisions of 1900, 1910, and 1920 were carried out under his leadership.
- As Secretary-General of the International Conference, he sent out the provisional revision for 1920 to more than 500 people, asking for comments.
- His death in 1922 left the International Conference without a guiding hand.

***At the 1923 session of the International Statistical Institute, Michel Huber, Bertillon's successor in France, recognized this lack of leadership and introduced a resolution for the International Statistical Institute to renew its stand of 1893 in regard to the International Classification of Causes of Death and to cooperate with other international organizations in preparation for subsequent revisions. The Health Organization of the League of Nations had also taken an active interest in vital statistics and appointed a Commission of Statistical Experts to study the classification of diseases and causes of death, as well as other problems in the field of medical statistics. E. Roesle, Chief of the Medical Statistical Service of the German Health Bureau and a member of the Commission of Expert Statisticians, prepared a monograph that listed the expansion in the rubrics of the 1920 International List of Causes of Death that would be required if the classification was to be used in the tabulation of statistics of morbidity. This careful study was published by the Health Organization of the League of Nations in 1928 (15). In order to coordinate the work of both agencies, an international commission, known as the "Mixed Commission", was created with an equal number of representatives from the International Statistical Institute and the Health Organization of the League of Nations. This Commission drafted the proposals for the Fourth (1929) and the Fifth (1938) revisions of the International List of Causes of Death.*[2]

The Decennial Revision

1 The Fifth Decennial Revision Conference

- *The Fifth International Conference for the Revision of the International List of Causes of Death, like the preceding conferences, was convened by the* ***Government of France and was held in Paris*** *in* ***October 1938***.
- The Conference approved three lists:
  - ✓*a detailed list of 200 titles*
  - ✓*an intermediate list of 87 titles*

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✓ *And an abridged list of 44 titles*.

1 Sixth Revision of the International

Lists the International Health Conference held in ***New York City in June and July 1946*** (*19*) entrusted the Interim Commission of the World Health Organization with the responsibility of:

> reviewing the existing machinery and of undertaking such preparatory work as may be necessary in connection with:
>
> (i) The next decennial revision of "The International Lists of Causes of Death" (including the lists adopted under the International Agreement of 1934, relating to Statistics of Causes of Death); and
>
> (ii) the establishment of International Lists of Causes of Morbidity

To meet this responsibility, the Interim Commission appointed the Expert Committee for the Preparation of the Sixth Decennial Revision of the International Lists of Diseases and Causes of Death.

This Committee, taking full account of prevailing opinion concerning morbidity and mortality classification, reviewed and revised the above-mentioned proposed classification which had been prepared by the United States Committee on Joint Causes of Death.

The resulting classification was circulated to national governments preparing morbidity and mortality statistics for comments and suggestions under the title, *International Classification of Diseases, Injuries, and Causes of Death.* The Expert Committee considered the replies and prepared a revised version incorporating such changes as appeared to improve the utility and acceptability of the classification. The Committee also compiled a list of diagnostic terms to appear under each title of the classification. Furthermore, a subcommittee was appointed to prepare a comprehensive alphabetical index of diagnostic statements classified to the appropriate category of the classification.

The Committee also considered the structure and uses of special lists of causes for tabulation and publication of morbidity and mortality statistics and studied other problems related to the international comparability of mortality statistics, such as form of medical certificate and rules for classification.

The International Conference for the Sixth Revision of the International Lists of Diseases and Causes of Death was convened in Paris from 26 to 30 April 1948 by the Government of France under the terms of the agreement signed at the close of the Fifth Revision Conference in 1938. Its secretariat was entrusted jointly to the competent French authorities and to the World Health Organization, which had carried out the preparatory work under the terms of the arrangement concluded by the governments represented at the International Health Conference in 1946 (*19*).

The Conference adopted the classification prepared by the Expert Committee as the Sixth Revision of the International Lists (*20*). It also considered other proposals of the Expert Committee concerning the compilation, tabulation and publication of morbidity and mortality statistics. The Conference approved the International Form of Medical Certificate of Cause of Death, accepted the underlying cause of death as the main cause to be tabulated, and endorsed the rules for selecting the underlying cause of death as well as the special lists for tabulation of morbidity and mortality data. It further recommended that the World Health Assembly should adopt regulations under *Article 21*(*b*) of the WHO Constitution to guide Member

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States in compiling morbidity and mortality statistics in accordance with the International Statistical Classification.

In 1948, the First World Health Assembly endorsed the report of the Sixth Revision Conference and adopted World Health Organization Regulations No. 1, prepared on the basis of the recommendations of the Conference. The International Classification, including the Tabular List of Inclusions defining the content of the categories, was incorporated, together with the form of the medical certificate of cause of death, the rules for classification and the special lists for tabulation, into the *Manual of the International Statistical Classification of Diseases, Injuries, and Causes of Death* (*21*). The Manual consisted of two volumes, Volume 2 being an alphabetical index of diagnostic terms coded to the appropriate categories.The Sixth Decennial Revision Conference marked the beginning of a new era in international vital and health statistics. Apart from approving a comprehensive list for both mortality and morbidity and agreeing on international rules for selecting the underlying cause of death, it recommended the adoption of a comprehensive programme of international cooperation in the field of vital and health statistics. An important item in this programme was the recommendation that governments establish national committees on vital and health statistics to coordinate the statistical activities in the country, and to serve as a link between the national statistical institutions and the World Health Organization. It was further envisaged that such national committees would, either singly or in cooperation with other national committees, study statistical problems of public health importance and make the results of their investigations available to WHO.

6.7 The Seventh and Eighth Revisions

The International Conference for the Seventh Revision of the International Classification of Diseases was held in Paris under the auspices of WHO in February 1955 (*22*). In accordance with a recommendation of the WHO Expert Committee on Health Statistics, this revision was limited to essential changes and amendments of errors and inconsistencies (*23*).

The Eighth Revision Conference convened by WHO met in Geneva, from 6 to 12 July 1965 (*24*). This revision was more radical than the Seventh but left unchanged the basic structure of the Classification and the general philosophy of classifying diseases, whenever possible, according to their etiology rather than a particular manifestation.

During the years that the Seventh and Eighth Revisions of the ICD were in force, the use of the ICD for indexing hospital medical records increased rapidly and some countries prepared national adaptations which provided the additional detail needed for this application of the ICD.

6.8 The Ninth Revision

The International Conference for the Ninth Revision of the International Classification of Diseases, convened by WHO, met in Geneva from 30 September to 6 October 1975 (*25*). In the discussions leading up to the conference, it had originally been intended that there should be little change other than updating of the classification. This was mainly because of the expense of adapting data-processing systems each time the classification was revised. There had been an enormous growth of interest in the ICD and ways had to be found of responding to this, partly by modifying the classification itself and partly by introducing special coding provisions. A number of representations were made by specialist bodies which had become interested in using the ICD for their own statistics. Some subject areas in the classification were regarded as inappropriately arranged and

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there was considerable pressure for more detail and for adaptation of the classification to make it more relevant for the evaluation of medical care, by classifying conditions to the chapters concerned with the part of the body affected rather than to those dealing with the underlying generalized disease. At the other end of the scale, there were representations from countries and areas where a detailed and sophisticated classification was irrelevant, but which nevertheless needed a classification based on the ICD in order to assess their progress in health care and in the control of disease.

The final proposals presented to and accepted by the Conference retained the basic structure of the ICD, although with much additional detail at the level of the four-digit subcategories, and some optional five-digit subdivisions. For the benefit of users not requiring such detail, care was taken to ensure that the categories at the three-digit level were appropriate.

For the benefit of users wishing to produce statistics and indexes oriented towards medical care, the Ninth Revision included an optional alternative method of classifying diagnostic statements, including information about both an underlying general disease and a manifestation in a particular organ or site. This system became known as the dagger and asterisk system and is retained in the Tenth Revision. A number of other technical innovations were included in the Ninth Revision, aimed at increasing its flexibility for use in a variety of situations.

The Twenty-ninth World Health Assembly, noting the recommendations of the International Conference for the Ninth Revision of the International Classification of Diseases, approved the publication, for trial purposes, of supplementary classifications of Impairments and Handicaps and of Procedures in Medicine as supplements to, but not as integral parts of, the International Classification of Diseases. The Conference also made recommendations on a number of related technical subjects: coding rules for mortality were amended slightly and rules for the selection of a single cause for tabulation of morbidity were introduced for the first time; definitions and recommendations for statistics in the field of perinatal mortality were amended and extended and a certificate of causes of perinatal death was recommended; countries were encouraged to do further work on multiple-condition coding and analysis, but no formal methods were recommended; and a new basic tabulation list was produced.

6.9 Preparations for the Tenth Revision

Even before the Conference for the Ninth Revision, WHO had been preparing for the Tenth Revision. It had been realized that the great expansion in the use of the ICD necessitated a thorough rethinking of its structure and an effort to devise a stable and flexible classification, which should not require fundamental revision for many years to come. The WHO Collaborating Centres for Classification of Diseases (see Volume 1) were consequently called upon to experiment with models of alternative structures for ICD-10.

It had also become clear that the established ten-year interval between revisions was too short. Work on the revision process had to start before the current version of the ICD had been in use long enough to be thoroughly evaluated, mainly because the necessity to consult so many countries and organizations made the process a very lengthy one. The Director-General of WHO therefore wrote to the Member States and obtained their agreement to postpone until 1989 the Tenth Revision Conference, which was originally scheduled for 1985 and to delay the introduction of the Tenth Revision which would have been due in 1989. In addition to permitting experimentation with alternative models for the structure of the ICD, this allowed time for the evaluation of ICD-9, for example through meetings

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organized by some of the WHO Regional Offices and through a survey organized at headquarters.

An extensive programme of work followed which culminated in the Tenth Revision of the ICD and is described in the Report of the International Conference for the Tenth Revision of the International Classification of Diseases, reproduced in Volume 1.[3]

1) EXPLAIN TYPE OF CLASSIFICATION OF DISEASE OPERATION & PROCEDURE

2) EXPLAIN PURPOSE OF DISEASE CLASSIFICATION OPERATION & PROCEDURE

A. PURPOSE OF DISEASE CLASSIFICATION

3) LIST & DESCRIBE The 22 MAJOR TYPE OF ICD

A. LIST OF 22 MAJOR CHAPTERS OF ICD

Tabular list of inclusions and four-character subcategories

I	Certain infectious and parasitic diseases
II	Neoplasms
III	Diseases of the blood and blood-forming organs and certain disorders involving the immune mechanism
IV	Endocrine, nutritional and metabolic diseases
V	Mental and behavioural disorders
VI	Diseases of the nervous system
VII	Diseases of the eye and adnexa
VIII	Diseases of the ear and mastoid process
IX	Diseases of the circulatory system
X	Diseases of the respiratory system
XI	Diseases of the digestive system
XII	Diseases of the skin and subcutaneous tissue
XIII	Diseases of the musculoskeletal system and connective tissue
XIV	Diseases of the genitourinary system
XV	Pregnancy, childbirth and the puerperium
XVI	Certain conditions originating in the perinatal period
XVII	Congenital malformations, deformations and chromosomal abnormalities
XVIII	Symptoms, signs and abnormal clinical and laboratory findings not elsewhere classified

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XIX Injury, poisoning and certain other consequences of external causes
XX External causes of morbidity and mortality
XXI Factors influencing health status and contact with health services[4]

B. ADDITIONAL CHAPTER

4) DESCRIBE THE ARRANGEMENT OF ICD & ICPM

A. CATEGORY CHARACTER

The basic structure and principles of classification of the ICD

- Volumes

✓ ICD-10 comprises three volumes:

➢ *VOLUME 1 CONTAINS THE MAIN CLASSIFICATIONS;*

➢ *VOLUME 2 PROVIDES GUIDANCE TO USERS OF THE ICD; AND*

✓ **Introduction**
✓ **Description of the International Statistical Classification of Diseases and Related Health Problems**
✓ **How to use the ICD**
✓ **Rules and guidelines for mortality and morbidity coding**
✓ **Statistical presentation**
✓ **History of the development of the ICD**
✓ **Appendices**

➢ *VOLUME 3 IS THE ALPHABETICAL INDEX TO THE CLASSIFICATION.* [5]

➢ **Arrangement of the Alphabetical Index**

➢ Volume 3 is divided into three sections as follows:

➢ • Section I lists all the terms classifiable to Chapters I-XIX and Chapter XXI, except drugs and other chemicals.

➢ • Section II is the index of external causes of morbidity and mortality and contains all the terms classifiable to Chapter XX, except drugs and other chemicals.

4 Extracted from ICD-10 Second Edition, 2005, 1: Tabular List.
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- • Section III, the Table of Drugs and Chemicals, lists for each substance the codes for poisonings and adverse effects of drugs classifiable to Chapter XIX, and the Chapter XX codes that indicate whether the poisoning was accidental, deliberate (self-harm), undetermined, or an adverse effect of a correct substance properly administered.

3.2.2 Structure

- The Index contains "lead terms", positioned to the far left of the column, with other words ("modifiers" or "qualifiers") at different levels of indentation under them. In Section I, these indented modifiers or qualifiers are usually varieties, sites or circumstances that affect coding; in Section II they indicate different types of accident or occurrence, vehicles involved, etc. Modifiers that do not affect coding appear in parentheses after the condition. [6]

- Chapters

- ✓ The classification is divided into 21 chapters. The first character of the ICD code is a letter, and each letter is associated with a particular chapter, except for the letter D, which is used in both Chapter II, Neoplasms, and Chapter III, Diseases of the blood and blood-forming organs and certain disorders involving the immune mechanism, and the letter H, which is used in both Chapter VII, Diseases of the eye and adnexa and Chapter VIII, Diseases of the ear and mastoid process. Four chapters (Chapters I, II, XIX, and XX) use more than one letter in the first position of their codes.
- ✓ Each chapter contains sufficient three-character categories to cover its content; not all available codes are used, allowing space for future revision and expansion.
- ✓ Chapters I to XVII relate to diseases and other morbid conditions, and Chapter XIX to injuries, poisoning and certain other consequences of external causes. The remaining chapters complete the range of subject matter nowadays included in diagnostic data. Chapter XVIII covers Symptoms, signs and abnormal clinical and laboratory findings, not elsewhere classified. Chapter XX, External causes of morbidity and mortality, was traditionally used to classify causes of injury and poisoning, but, since the Ninth Revision, has also provided for any recorded external cause of diseases and other morbid conditions. Finally, Chapter XXI, Factors influencing health status and contact with health services, is intended for the classification of data explaining the reason for contact with health care services of a person not currently sick, or the circumstances in which the patient is receiving care at that particular time or otherwise having some bearing on that person's care. [7]

- Blocks of categories

- ✓ The chapters are subdivided into homogeneous "blocks" of three-character categories. In Chapter I, the block titles reflect two axes of classification - mode of transmission and broad group of infecting organisms. In Chapter II, the first axis is the behaviour of the neoplasm; within behaviour, the axis is mainly by site, although a few three-character categories are provided for important morphological types (e.g. leukaemias, lymphomas, melanomas, mesotheliomas, Kaposi's sarcoma). The range of categories is given in parentheses after each block title. [8]

- Three-character categories

- ✓ Within each block, some of the three-character categories are for single conditions, selected because of their frequency, severity or susceptibility to public health intervention, while others are for groups of diseases with some common characteristic. There is usually provision for "other" conditions to be classified, allowing many different but rarer conditions, as well as "unspecified" conditions, to be included. [9]

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6 Extracted from ICD-10 Second Edition, 2005, 3. How to use the ICD.

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- Four-character subcategories

✓ Although not mandatory for reporting at the international level, most of the three-character categories are subdivided by means of a fourth, numeric character after a decimal point, allowing up to ten subcategories. Where a three-character category is not subdivided, it is recommended that the letter "X" be used to fill the fourth position so that the codes are of a standard length for data-processing.

✓ The four-character subcategories are used in whatever way is most appropriate, identifying, for example, different sites or varieties if the threecharacter category is for a single disease, or individual diseases if the threecharacter category is for a group of conditions.

✓ The fourth character .8 is generally used for "other" conditions belonging to the three-character category, and .9 is mostly used to convey the same meaning as the three-character category title, without adding any additional information.

✓ When the same fourth-character subdivisions apply to a range of three-character categories, they are listed once only, at the start of the range. A note at each of the relevant categories indicates where the details are to be found. For example, categories O03-O06, for different types of abortion, have common fourth characters relating to associated complications (see Volume 1). [10]

- Supplementary subdivisions for use at the fifth or subsequent character level

- The fifth and subsequent character levels are usually subclassifications along a different axis from the fourth character. They are found in:
- Chapter XIII - subdivisions by anatomical site
- Chapter XIX - subdivisions to indicate open and closed fractures as well as intracranial, intrathoracic and intra-abdominal injuries with and without open wound
- Chapter XX - subdivisions to indicate the type of activity being undertaken at the time of the event. [11]

- The unused "U" codes[12]

Codes U00-U49 are to be used for the provisional assignment of new diseases of uncertain etiology. Codes U50-U99 may be used in research, e.g. when testing an alternative subclassification for a special project.[13]

B. LAYOUT STRUCTURE OF ICD

C. LAYOUT STRUCTURE OF ICPM

5) EXPLAIN THE PROCEDURE OF INDEXING & CODING

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1. CODING PROCEDURE

2. 3.3 Basic coding guidelines

3. The Alphabetical Index contains many terms not included in Volume 1, and coding requires that both the Index and the Tabular List should be consulted before a code is assigned.

4. Before attempting to code, the coder needs to know the principles of classification and coding and to have carried out practical exercises.

5. The following is a simple guide intended to assist the occasional user of the ICD.

6. 1. Identify the type of statement to be coded and refer to the appropriate section of the Alphabetical Index. (If the statement is a disease or injury or other condition classifiable to Chapters I-XIX or XXI, consult Section I of the Index. If the statement is the external cause of an injury or other event classifiable to Chapter XX, consult Section II.)

7. 2. Locate the lead term. For diseases and injuries this is usually a noun for the pathological condition. However, some conditions expressed as adjectives or eponyms are included in the Index as lead terms.

8. 3. Read and be guided by any note that appears under the lead term.

9. 4. Read any terms enclosed in parentheses after the lead term (these modifiers do not affect the code number), as well as any terms indented under the lead term (these modifiers may affect the code number), until all the words in the diagnostic expression have been accounted for.

10. 5. Follow carefully any cross-references ("see" and "see also") found in the Index.

11. 6. Refer to the tabular list to verify the suitability of the code number selected. Note that a three-character code in the Index with a dash in the fourth position means that there is a fourth character to be found in Volume 1. Further subdivisions to be used in a supplementary character position are not indexed and, if used, must be located in Volume 1.

12. 7. Be guided by any inclusion or exclusion terms under the selected code or under the chapter, block or category heading.

13. 8. Assign the code.

14. Specific guidelines for the selection of the cause or condition to be coded, and for coding the condition selected, are given in Section 4.[14]

15. THINGS TO CONCEDE WHEN CODING

16. Indexing PROCEDURE

6) DESCRIBE ABBREVIATION, CONVENTION, PUNCTUATION AS USED IN ICD & ICPM

7) DEFINE & DESCRIBE THE DIAGNOSTIC INDEXING

14 Extracted from ICD-10 Second Edition, 2005, 3. How to use the ICD.

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- ***Definition of terms***
    - ***Indexing***
    - ***Disease***
    - ***Healthy***
    - ***Medicine***
    - ***Data***
    - ***Information***
    - ***Morbidity***
    - ***Mortality***
    - ***Fatality***
    - ***Code***
    - ***Data/ information classification***
    - ***System***
- ***List of system of data classification***
- ***Record to be indexed***
- ***Material to maintain a Diagnostic indexing***
    - ***Cards***
        - ✓ ***Layout of diagnostic index card***
        - ✓ ***Procedure for filling card***
    - ***Register***
        - ✓ ***Layout of diagnostic indexing card***
    - ***Computer(e-coding)***
    - ***Loose leaf binder***
        - ✓ ***Layout of diagnostic indexing loose leaf binder***

- ***Use/ purpose of data classification***
- ***Procedure for indexing***
- ***Summary of diagnostic index data***
- ***Procedure for making summary in diagnostic indexing***
- ***Coding & indexing in icpm***
- ***Levels of data classification***

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8) DEMONSTRATE ABILITY CODE & INDEXING & MEDICAL PROCEDURE IN ICD & ICPM

9) DEMONSTRATE ABILITY TO CREATE MANUAL AND ELECTRIC INDEXING

10) DESCRIBE VARIOUS MEDICAL TERM USED CODING & CLASSIFICATION OF DISEASE

Inclusion terms

Within the three- and four-character rubrics[1], there are usually listed a number of other diagnostic terms. These are known as "inclusion terms" and are given, in addition to the title, as examples of the diagnostic statements to be classified to that rubric. They may refer to different conditions or be synonyms. They are not a subclassification of the rubric.

Inclusion terms are listed primarily as a guide to the content of the rubrics. Many of the items listed relate to important or common terms belonging to the rubric. Others are borderline conditions or sites listed to distinguish the boundary between one subcategory and another. The lists of inclusion terms are by no means exhaustive and alternative names of diagnostic entities are included in the Alphabetical Index, which should be referred to first when coding a given diagnostic statement.

It is sometimes necessary to read inclusion terms in conjunction with titles. This usually occurs when the inclusion terms are elaborating lists of sites or pharmaceutical products, where appropriate words from the title (e.g. "malignant neoplasm of ...", "injury to ...", "poisoning by ...") need to be understood.

> General diagnostic descriptions common to a range of categories, or to all the subcategories in a three-character category, are to be found in notes headed "Includes", immediately following a chapter, block or category title.

[1] In the context of the ICD, "rubric" denotes either a three-character category or a four-character subcategory.

Exclusion terms

Certain rubrics contain lists of conditions preceded by the word "Excludes". These are terms which, although the rubric title might suggest that they were to be classified there, are in fact classified elsewhere. An example of this is in category A46, "Erysipelas", where postpartum or puerperal erysipelas is excluded. Following each excluded term, in parentheses, is the category or subcategory code elsewhere in the classification to which the excluded term should be allocated.

General exclusions for a range of categories or for all subcategories in a three-character category are to be found in notes headed "Excludes", immediately following a chapter, block or category title. [15]

The "dagger and asterisk" system

ICD-9 introduced a system, continued in ICD-10, whereby there are two codes for diagnostic statements containing information about both an underlying generalized disease and a manifestation in a particular organ or site which is a clinical problem in its own right.

15.

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The primary code is for the underlying disease and is marked with a dagger (†); an optional additional code for the manifestation is marked with an asterisk (*). This convention was provided because coding to underlying disease alone was often unsatisfactory for compiling statistics relating to particular specialties, where there was a desire to see the condition classified to the relevant chapter for the manifestation when it was the reason for medical care.

While the dagger and asterisk system provides alternative classifications for the presentation of statistics, it is a principle of the ICD that the dagger code is the primary code and must always be used. Provision should be made for the asterisk code to be used in addition if the alternative method of presentation may also be required. For coding, the asterisk code must never be used alone. Statistics incorporating the dagger codes conform with the traditional classification for presenting data on mortality and morbidity and other aspects of medical care.

Asterisk codes appear as three-character categories. There are separate categories for the same conditions occurring when a particular disease is not specified as the underlying cause. For example, categories G20 and G21 are for forms of Parkinsonism that are not manifestations of other diseases assigned elsewhere, while category G22* is for "Parkinsonism in diseases classified elsewhere". Corresponding dagger codes are given for conditions mentioned in asterisk categories; for example, for Syphilitic parkinsonism in G22*, the dagger code is A52.1†.

Some dagger codes appear in special dagger categories. More often, however, the dagger code for dual-element diagnoses and unmarked codes for single-element conditions may be derived from the same category or subcategory.

The areas of the classification where the dagger and asterisk system operates are limited; there are 83 special asterisk categories throughout the classification, which are listed at the start of the relevant chapters.

Rubrics in which dagger-marked terms appear may take one of three different forms:

(i) If the symbol (†) and the alternative asterisk code both appear in the rubric heading, all terms classifiable to that rubric are subject to dual classification and all have the same alternative code, e.g.

A17.0† Tuberculous meningitis (G01*)
Tuberculosis of meninges (cerebral) (spinal)
Tuberculous leptomeningitis

(ii) If the symbol appears in the rubric heading but the alternative asterisk code does not, all terms classifiable to that rubric are subject to dual classification but they have different alternative codes (which are listed for each term), e.g.

A18.1† Tuberculosis of genitourinary system

Tuberculosis of:
- bladder (N33.0*)
- cervix (N74.0*)
- kidney (N29.1*)
- male genital organs (N51.-*)
- ureter (N29.1*)

Tuberculous female pelvic inflammatory disease (N74.1*)

(iii) If neither the symbol nor the alternative code appear in the title, the rubric as a whole is not subject to dual classification but individual inclusion terms may be; if so, these terms will be marked with the symbol and their alternative codes given, e.g.

A54.8 Other gonococcal infections Gonococcal:

...
- peritonitis† (K67.1*)
- pneumonia† (J17.0*)
- septicaemia
- skin lesions

Other optional dual coding

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There are certain situations, other than in the dagger and asterisk system, that permit two ICD codes to be used to describe fully a person's condition. The note in the tabular list, “Use additional code, if desired ...”, identifies many of these situations. The additional codes would be used only in special tabulations.

These are:

(i) for local infections, classifiable to the “body systems” chapters, codes from Chapter I may be added to identify the infecting organism, where this information does not appear in the title of the rubric. A block of categories, B95-B97, is provided for this purpose in Chapter I.

(ii) for neoplasms with functional activity. To the code from Chapter II may be added the appropriate code from Chapter IV to indicate the type of functional activity.

(iii) for neoplasms, the morphology code from Chapter I, although not part of the main ICD, may be added to the Chapter II code to identify the morphological type of the tumour.

(iv) for conditions classifiable to F00-F09 (Organic, including symptomatic, mental disorders) in Chapter V, where a code from another chapter may be added to indicate the cause, i.e. the underlying disease, injury or other insult to the brain.

(v) where a condition is caused by a toxic agent, a code from Chapter XX may be added to identify that agent.

(vi) where two codes can be used to describe an injury, poisoning or other adverse effect: a code from Chapter XIX, which describes the nature of the injury, and a code from Chapter XX, which describes the cause. The choice as to which code should be the additional code depends upon the purpose for which the data are being collected. (See introduction to Chapter XX, of Volume 1.)

3.1.4 Conventions used in the tabular list

In listing inclusion and exclusion terms in the tabular list, the ICD employs some special conventions relating to the use of parentheses, square brackets, colons, braces, the abbreviation “NOS”, the phrase “not elsewhere classified” (NEC), and the word “and” in titles. These need to be clearly understood both by coders and by anyone wishing to interpret statistics based on the ICD.

Parentheses ( )

Parentheses are used in Volume 1 in four important situations.

(a) Parentheses are used to enclose supplementary words, which may follow a diagnostic term without affecting the code number to which the words outside the parentheses would be assigned. For example, in I10 the inclusion term, “Hypertension (arterial) (benign) (essential) (malignant) (primary) (systemic)”, implies that I10 is the code number for the word “Hypertension” alone or when qualified by any, or any combination, of the words in parentheses.

(b) Parentheses are also used to enclose the code to which an exclusion term refers. For example, H01.0 Blepharitis Excludes: blepharoconjunctivitis (H10.5).

(c) Another use of parentheses is in the block titles, to enclose the threecharacter codes of categories included in that block.

(d) The last use of parentheses was incorporated in the Ninth Revision and is related to the dagger and asterisk system. Parentheses are used to enclose the dagger code in an asterisk category or the asterisk code following a dagger term.

Square brackets [ ]

Square brackets are used:

(a) for enclosing synonyms, alternative words or explanatory phrases; for example,

A30 Leprosy [Hansen's disease];

(b) for referring to previous notes; for example,

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C00.8 Overlapping lesion of lip [See note 5 at the beginning of this chapter];

(c) for referring to a previously stated set of fourth character subdivisions common to a number of categories; for example,

K27 Peptic ulcer, site unspecified [See before K25 for subdivisions].

Colon :

A colon is used in listings of inclusion and exclusion terms when the words that precede it are not complete terms for assignment to that rubric. They require one or more of the modifying or qualifying words indented under them before they can be assigned to the rubric. For example, in K36, "Other appendicitis", the diagnosis "appendicitis" is to be classified there only if qualified by the words "chronic" or "recurrent".

Brace }

A brace is used in listings of inclusion and exclusion terms to indicate that neither the words that precede it nor the words after it are complete terms. Any of the terms before the brace should be qualified by one or more of the terms that follow it. For example:

O71.6	Obstetric damage to pelvic joints and ligaments		
	Avulsion of inner symphyseal cartilage	}	
	Damage to coccyx	}	obstetric
	Traumatic separation of symphysis (pubis)	}	

"NOS"

The letters NOS are an abbreviation for "not otherwise specified", implying "unspecified" or "unqualified". Sometimes an unqualified term is nevertheless classified to a rubric for a more specific type of the condition. This is because, in medical terminology, the most common form of a condition is often known by the name of the condition itself and only the less common types are qualified. For example, "mitral stenosis" is commonly used to mean "rheumatic mitral stenosis". These inbuilt assumptions have to be taken into account in order to avoid incorrect classification. Careful inspection of inclusion terms will reveal where an assumption of cause has been made; coders should be careful not to code a term as unqualified unless it is quite clear that no information is available that would permit a more specific assignment elsewhere. Similarly, in interpreting statistics based on the ICD, some conditions assigned to an apparently specified category will not have been so specified on the record that was coded. When comparing trends over time and interpreting statistics, it is important to be aware that assumptions may change from one revision of the ICD to another. For example, before the Eighth Revision, an unqualified aortic aneurysm was assumed to be due to syphilis.

"Not elsewhere classified"

The words "not elsewhere classified", when used in a three-character category title, serve as a warning that certain specified variants of the listed conditions may appear in other parts of the classification. For example:

J16 Pneumonia due to other infectious organisms, not elsewhere classified

This category includes J16.0 Chlamydial pneumonia and J16.8 Pneumonia due to other specified infectious organisms. Many other categories are provided in Chapter X (for example, J10-J15) and other chapters (for example, P23.- Congenital pneumonia) for pneumonias due to specified infectious organisms. J18 Pneumonia, organism unspecified, accommodates pneumonias for which the infectious agent is not stated.

"And" in titles

"And" stands for "and/or". For example, in the rubric A18.0, Tuberculosis of bones and joints, are to be classified cases of "tuberculosis of bones", "tuberculosis of joints" and "tuberculosis of bones and joints".

Point dash.-

In some cases, the fourth character of a subcategory code is replaced by a dash, e.g.

G03 Meningitis due to other and unspecified causes,
Excludes: meningoencephalitis (G04.-)

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This indicates to the coder that a fourth character exists and should be sought in the appropriate category. This convention is used in both the tabular list and the alphabetical index.

3.1.5 Categories with common characteristics

For quality control it is useful to introduce programmed checks into the computer system. The following groups of categories are provided as a basis for such checks on internal consistency, grouped according to the special characteristic that unites them.

Asterisk categories

The following asterisk categories are not to be used alone; they must always be used in addition to a dagger code:

D63*, D77*, E35*, E90*, F00*, F02*, G01*, G02*, G05*, G07*, G13*, G22*, G26*, G32*, G46*, G53*, G55*, G59*, G63*, G73*, G94*, G99*, H03*, H06*, H13*, H19*, H22*, H28*, H32*, H36*, H42*, H45*, H48*, H58*, H62*, H67*, H75*, H82*, H94*, I32*, I39*, I41*, I43*, I52*, I68*, I79*, I98*, J17*, J91*, J99*, K23*, K67*, K77*, K87*, K93*, L14*, L45*, L54*, L62*, L86*, L99*, M01*, M03*, M07*, M09*, M14*, M36*, M49*, M63*, M68*, M73*, M82*, M90*, N08*, N16*, N22*, N29*, N33*, N37*, N51*, N74*, N77*, P75* [16]

3.2.3 Code numbers

The code numbers that follow the terms refer to the categories and subcategories to which the terms should be classified. If the code has only three characters, it can be assumed that the category has not been subdivided. In most instances where the category has been subdivided, the code number in the Index will give the fourth character. A dash in the fourth position (e.g. O03.-) means that the category has been subdivided and that the fourth character can be found by referring to the tabular list. If the dagger and asterisk system applies to the term, both codes are given.

3.2.4 Conventions

Parentheses

Parentheses are used in the Index in the same way as in Volume 1, i.e. to enclose modifiers.

"NEC"

NEC (not elsewhere classified) indicates that specified variants of the listed condition are classified elsewhere, and that, where appropriate, a more precise term should be looked for in the Index.

Cross-references

Cross-references are used to avoid unnecessary duplication of terms in the Index. The word "see" requires the coder to refer to the other term; "see also" directs the coder to refer elsewhere in the Index if the statement being coded contains other information that is not found indented under the term to which "see also" is attached. [17]

11) DIFFERENTIATE BETWEEN PRIMARY & SECONDARY DIAGNOSTIC & DUAL CLASSIFICATION

12) DISPLAY TIREDNESS, LEGIBILITY & ACCURACY IN COMPLETING INDEXING

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13) **DEMONSTRATE ABILITY TO EDIT AUDIT CASE RECORD**

14) **DEMONSTRATE ABILITY TO GENERATE ANALYZE & INTERPRETED THE INFORMATION**

15) **DESCRIBE THE PROCEDURE IN DATA SECURITY & CONFIDENTIALITY**

16) CITIFICATION

International form of medical certificate of cause of death

- The above principle can be applied ***uniformly by using the medical certification form recommended by the World Health Assembly***.
- It is the *responsibility of the* ***medical practitioner signing the death certificate to indicate which;***
  - ✓ *Morbid conditions led directly to death*
  - ✓ *And to state any antecedent conditions giving rise to this cause.*

The medical certificate shown below is designed to facilitate the selection of the underlying cause of death when two or more causes are recorded.

Part I of the form is for diseases related to the train of events leading directly to death, and Part II is for unrelated but contributory conditions.

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INTERNATIONAL FORM OF MEDICAL CERTIFICATE OF CAUSE OF DEATH

Cause of death		Approximate interval between onset and death
I		
Disease or condition directly leading to death*	(a)	. . . . .
	due to (or as a consequence of)	
Antecedent causes Morbid conditions, if any, giving rise to the above cause, stating the underlying condition last	(b)	. . . . .
	due to (or as a consequence of)	
	(c)	. . . . .
	due to (or as a consequence of)	
	(d)	. . . . .
II		
Other significant conditions contributing to the death, but not related to the disease or condition causing it	. . . . .	. . . . .
	. . . . .	. . . . .

**This does not mean the mode of dying, e.g. heart failure, respiratory failure. It means the disease, injury, or complication that caused death.*

The medical practitioner or other qualified certifier should use his or her clinical judgement in completing the medical certificate of cause of death. Automated systems must not include lists or other prompts to guide the certifier as these necessarily limit the range of diagnoses and therefore have an adverse effect on the accuracy and usefulness of the report.

In 1990, the Forty-third World Health Assembly adopted a recommendation that, where a need had been identified, countries should consider the possibility of an additional line, (d), in Part I of the certificate. However, countries may adopt, or continue to use, a certificate with only three lines in Part I where a fourth line is unnecessary, or where there are legal or other impediments to the adoption of the certificate shown above.

The condition recorded on the lowest used line of Part I of the certificate is usually the underlying cause of death used for tabulation. However, the procedures described in sections 4.1.4-4.1.5 may result in the selection of another condition as the underlying cause of death. To differentiate between these two possibilities, the expression originating antecedent cause (originating cause) will be used to refer to the condition proper to the last used line of Part I of the certificate, and the expression underlying cause of death will be used to identify the cause selected for tabulation.

If there is only one step in the chain of events, an entry at line I(a) is sufficient. If there is more than one step, the direct cause is entered at (a) and the originating

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antecedent cause is entered last, with any intervening cause entered on line (b) or on lines (b) and (c). An example of a death certificate with four steps in the chain of events leading directly to death is:

(a) Pulmonary embolism
(b) Pathological fracture
(c) Secondary carcinoma of femur
(d) Carcinoma of breast

Part II is for any other significant condition that contributed to the fatal outcome, but was not related to the disease or condition directly causing death.

After the words "due to (or as a consequence of)", which appear on the certificate, should be included not only the direct cause or pathological process, but also indirect causes, for example where an antecedent condition has predisposed to the direct cause by damage to tissues or impairment of function, even after a long interval.

Noting the approximate interval (minutes, hours, days, weeks, months or years) between the onset of each condition and the date of death helps the certifying doctor to establish the chain of events that led to the death, and is also useful subsequently in guiding the coder to choose the appropriate code.

In 1990, the World Health Assembly adopted a recommendation that countries should consider the inclusion on death certificates of questions about current pregnancy and pregnancy within one year preceding death. [18]

17) RULES AN D GUIDELINES FOR MORBIDITY AND MORTALITY

18) STATISTICAL PRESENTATION

Definition of terms

The immediate cause of disease

4.1.1 Causes of death

In 1967, the Twentieth World Health Assembly defined the causes of death to be entered on the medical certificate of cause of death as "all those diseases, morbid conditions or injuries which either resulted in or contributed to death and the circumstances of the accident or violence which produced any such injuries". The purpose of the definition is to ensure that all the relevant information is recorded and that the certifier does not select some conditions for entry and reject others. The definition does not include symptoms and modes of dying, such as heart failure or respiratory failure.

When only one cause of death is recorded, this cause is selected for tabulation. When more than one cause of death is recorded, selection should be made in accordance with the rules given in section 4.1.5. The rules are based on the concept of the underlying cause of death. [19]

Underlying cause of death

18 Extracted from ICD-10 Second Edition, 2005, 4. Rules and guidelines for mortality and morbidity coding.

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It was agreed by the Sixth Decennial International Revision Conference that the cause of death for primary tabulation should be designated the underlying cause of death. From the standpoint of prevention of death, it is necessary to break the chain of events or to effect a cure at some point. The most effective public health objective is to prevent the precipitating cause from operating. For this purpose, the underlying cause has been defined as "(a) the disease or injury which initiated the train of morbid events leading directly to death, or (b) the circumstances of the accident or violence which produced the fatal injury". [20]

Importance of underline cause of disease

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19 **4.1.4 Procedures for selection of the underlying cause of death for mortality tabulation**

When only one cause of death is reported, this cause is used for tabulation.

When more than one cause of death is recorded, the first step in selecting the underlying cause is to determine the originating antecedent cause proper to the lowest used line in Part I of the certificate by application of the General Principle or of selection rules 1, 2 and 3.

In some circumstances the ICD allows the originating cause to be superseded by one more suitable for expressing the underlying cause in tabulation. For example, there are some categories for combinations of conditions, or there may be overriding epidemiological reasons for giving precedence to other conditions on the certificate.

The next step therefore is to determine whether one or more of the modification rules A to F (see section 4.1.9), which deal with the above situations, apply. The resultant code number for tabulation is that of the underlying cause.

Where the originating antecedent cause is an injury or other effect of an external cause classified to Chapter XIX, the circumstances that gave rise to that condition should be selected as the underlying cause for tabulation and coded to V01-Y89. The code for the injury or effect may be used as an additional code.

4.1.5 Rules for selection of the originating antecedent cause

Sequence

The term "sequence" refers to two or more conditions entered on successive lines of Part I, each condition being an acceptable cause of the one entered on the line above it.

Example 1: I (a) Bleeding of oesophageal varices
(b) Portal hypertension
(c) Liver cirrhosis
(d) Hepatitis B

If there is more than one cause of death in a line of the certificate, it is possible to have more than one reported sequence. In the example below, four sequences are reported:

Example 2: I (a) Coma
(b) Myocardial infarction and cerebrovascular accident
(c) Atherosclerosis Hypertension

The sequences are:

- atherosclerosis (leading to) myocardial infarction (leading to) coma;
- atherosclerosis (leading to) cerebrovascular accident (leading to) coma;
- hypertension (leading to) myocardial infarction (leading to) coma;
- hypertension (leading to) cerebrovascular accident (leading to) coma;

General Principle

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Clinician role

Mortality tabulation- data management role

The General Principle states that when more than one condition is entered on the certificate, the condition entered alone on the lowest used line of Part I should be selected only if it could have given rise to all the conditions entered above it.

Selection rules

Rule 1. If the General Principle does not apply and there is a reported sequence terminating in the condition first entered on the certificate, select the originating cause of this sequence. If there is more than one sequence terminating in the condition mentioned first, select the originating cause of the first-mentioned sequence.

Rule 2. If there is no reported sequence terminating in the condition first entered on the certificate, select this first-mentioned condition.

Rule 3. If the condition selected by the General Principle or by Rule l or Rule 2 is obviously a direct consequence of another reported condition, whether in Part I or Part II, select this primary condition.

4.1.6 Some considerations on selection rules

In a properly completed certificate, the originating antecedent cause will have been entered alone on the lowest used line of Part I and the conditions, if any, that arose as a consequence of this initial cause will have been entered above it, one condition to a line in ascending causal order.

Example 3:	I	(a)	Uraemia
		(b)	Hydronephrosis
		(c)	Retention of urine
		(d)	Hypertrophy of prostate
Example 4:	I	(a)	Bronchopneumonia
		(b)	Chronic bronchitis
	II		Chronic myocarditis

In a properly completed certificate, therefore, the General Principle will apply. However, even if the certificate has not been properly completed, the General Principle may still apply provided that the condition entered alone on the lowest used line of Part I could have given rise to all the conditions above it, even though the conditions entered above it have not been entered in the correct causal order.

Example 5:	I	(a)	Generalized metastases	5	weeks
		(b)	Bronchopneumonia	3	days
		(c)	Lung cancer	11	months

The General Principle does not apply when more than one condition has been entered on the lowest used line of Part I, or if the single condition entered could not have given rise to all the conditions entered above it. Guidance on the acceptability of different sequences is given at the end of the rules, but it should be borne in mind that the medical certifier's statement reflects an informed opinion about the conditions leading to death and about their interrelationships, and should not be disregarded lightly.

Where the General Principle cannot be applied, clarification of the certificate should be sought from the certifier whenever possible, since the selection rules are somewhat arbitrary and may not always lead to a satisfactory

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selection of the underlying cause. Where further clarification cannot be obtained, however, the selection rules must be applied. Rule l is applicable only if there is a reported sequence, terminating in the condition first entered on the certificate. If such a sequence is not found, Rule 2 applies and the first-entered condition is selected.

The condition selected by the above rules may, however, be an obvious consequence of another condition that was not reported in a correct causal relationship with it, e.g. in Part II or on the same line in Part I. If so, Rule 3 applies and the originating primary condition is selected. It applies, however, only when there is no doubt about the causal relationship between the two conditions; it is not sufficient that a causal relationship between them would have been accepted if the certifier had reported it.

4.1.7 Examples of the General Principle and selection rules

General Principle

When more than one condition is entered on the certificate, select the condition entered alone on the lowest used line of Part I only if it could have given rise to all the conditions entered above it.

Example 6:

I (a) Abscess of lung
(b) Lobar pneumonia

Select lobar pneumonia (J18.1).

Example 7:

I (a) Hepatic failure
(b) Bile duct obstruction
(c) Carcinoma of head of pancreas

Select carcinoma of head of pancreas (C25.0).

Example 8:

I (a) Cerebral haemorrhage
(b) Hypertension
(c) Chronic pyelonephritis
(d) Prostatic adenoma

Select prostatic adenoma (N40).

Example 9:

I (a) Traumatic shock
(b) Multiple fractures
(c) Pedestrian hit by truck (traffic accident)

Select pedestrian hit by truck (V04.1).

Example 10:

I (a) Bronchopneumonia
II Secondary anaemia and chronic lymphatic leukaemia

Select bronchopneumonia. But Rule 3 also applies; see Example 26.

Rule 1

If the General Principle does not apply and there is a reported sequence terminating in the condition first entered on the certificate, select the originating cause of this sequence. If there is more than one sequence

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terminating in the condition mentioned first, select the originating cause of the first-mentioned sequence.

Example 11: I (a) Bronchopneumonia
(b) Cerebral infarction and hypertensive heart disease

Select cerebral infarction (I63.9). There are two reported sequences terminating in the condition first entered on the certificate; bronchopneumonia due to cerebral infarction, and bronchopneumonia due to hypertensive heart disease. The originating cause of the first-mentioned sequence is selected.

Example 12: I (a) Oesophageal varices and congestive heart failure
(b) Chronic rheumatic heart disease and cirrhosis of liver

Select cirrhosis of liver (K74.6). The sequence terminating in the condition first entered on the certificate is oesophageal varices due to cirrhosis of liver.

Example 13: I (a) Acute myocardial infarction
(b) Atherosclerotic heart disease
(c) Influenza

Select atherosclerotic heart disease. The reported sequence terminating in the condition first entered on the certificate is acute myocardial infarction due to atherosclerotic heart disease. But Modification Rule C also applies; see Example 45.

Example 14: I (a) Pericarditis
(b) Uraemia and pneumonia

Select uraemia. There are two reported sequences terminating in the condition first entered on the certificate: pericarditis due to uraemia and pericarditis due to pneumonia. The originating cause of the first-mentioned sequence is selected. But Modification Rule D also applies; see Example 60.

Example 15: I (a) Cerebral infarction and hypostatic pneumonia
(b) Hypertension and diabetes
(c) Atherosclerosis

Select atherosclerosis. There are two reported sequences terminating in the condition first entered on the certificate: cerebral infarction due to hypertension due to atherosclerosis and cerebral infarction due to diabetes. The originating cause of the first-mentioned sequence is selected. But Modification Rule C also applies; see Example 46.

Rule 2

If there is no reported sequence terminating in the condition first entered on the certificate, select this first-mentioned condition.

Example 16: I (a) Pernicious anaemia and gangrene of foot
(b) Atherosclerosis

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Select pernicious anaemia (D51.0). There is no reported sequence terminating in the first entered condition.

Example 17: I (a) Rheumatic and atherosclerotic heart disease

Select rheumatic heart disease (I09.9). There is no reported sequence; both conditions are on the same line.

Example 18: I (a) Fibrocystic disease of the pancreas
(b) Bronchitis and bronchiectasis

Select fibrocystic disease of the pancreas (E84.9). There is no reported sequence.

Example 19: I (a) Senility and hypostatic pneumonia
(b) Rheumatoid arthritis

Select senility. There is a reported sequence - hypostatic pneumonia due to rheumatoid arthritis - but it does not terminate in the condition first entered on the certificate. But Modification Rule A also applies; see Example 33.

Example 20: I (a) Bursitis and ulcerative colitis

Select bursitis. There is no reported sequence. But Modification Rule B also applies; see Example 41.

Example 21: I (a) Acute nephritis, scarlet fever.

Select acute nephritis. There is no reported sequence. But Rule 3 also applies; see Example 28.

Rule 3

If the condition selected by the General Principle or by Rule l or Rule 2 is obviously a direct consequence of another reported condition, whether in Part I or Part II, select this primary condition.

Assumed direct consequences of another condition

Kaposi's sarcoma, Burkitt's tumour and any other malignant neoplasm of lymphoid, haematopoietic and related tissue, classifiable to C46.- or C81-C96, should be considered to be a direct consequence of HIV disease, where this is reported. No such assumption should be made for other types of malignant neoplasm.

Any infectious disease classifiable to A00-B19, B25-B49, B58-B64, B99 or J12-J18 should be considered to be a direct consequence of reported HIV disease.

Certain postoperative complications (pneumonia (any type), haemorrhage, thrombophlebitis, embolism, thrombosis, septicaemia, cardiac arrest, renal failure (acute), aspiration, atelectasis and infarction) can be considered direct consequences of an operation, unless surgery was carried out four weeks or more before death.

Any pneumonia in J12-J18 should be considered an obvious consequence of conditions that impair the immune system. Pneumonia in J18.0 and J18.2-J18.9 should be considered an obvious consequence of wasting diseases (such as malignant neoplasm and malnutrition) and diseases causing paralysis (such as cerebral haemorrhage or thrombosis), as well as serious respiratory conditions, communicable diseases, and serious injuries. Pneumonia in

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J18.0 and J18.2-J18.9, J69.0, and J69.8 should also be considered an obvious consequence of conditions that affect the process of swallowing.

Any disease described or qualified as “embolic” may be assumed to be a direct consequence of venous thrombosis, phlebitis or thrombophlebitis, valvular heart disease, atrial fibrillation, childbirth or any operation.

Any disease described as secondary should be assumed to be a direct consequence of the most probable primary cause entered on the certificate.

Secondary or unspecified anaemia, malnutrition, marasmus or cachexia may be assumed to be a consequence of any malignant neoplasm.

Any pyelonephritis may be assumed to be a consequence of urinary obstruction from conditions such as hyperplasia of prostate or ureteral stenosis.

Nephritic syndrome may be assumed to be a consequence of any streptococcal infection (scarlet fever, streptococcal sore throat, etc.).

Dehydration may be assumed to be a consequence of any intestinal infectious disease.

An operation on a given organ should be considered a direct consequence of any surgical condition (such as malignant tumour or injury) of the same organ reported anywhere on the certificate.

Example 22:

I (a) Kaposi's sarcoma
II AIDS

Select HIV disease resulting in Kaposi's sarcoma (B21.0)

Example 23:

I (a) Cancer of ovary
II HIV disease

Select malignant neoplasm of ovary (C56)

Example 24:

I (a) Tuberculosis
II HIV disease

Select HIV disease resulting in mycobacterial infection (B20.0)

Example 25:

I (a) Cerebral toxoplasmosis and herpes zoster
(b) Burkitt's lymphoma, HIV disease

Select HIV disease resulting in multiple diseases classified elsewhere (B22.7). Cerebral toxoplasmosis, selected by Rule 2, can be considered a direct consequence of HIV disease.

Example 26:

I (a) Bronchopneumonia
II Secondary anaemia and chronic lymphatic leukaemia

Select chronic lymphatic leukaemia (C91.1). Bronchopneumonia, selected by the General Principle (see Example 10), and secondary anaemia can both be considered direct sequels of chronic lymphatic leukaemia.

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Example 27:

I (a) Cerebral haemorrhage
(b) Hypertension
(c) Chronic pyelonephritis and prostatic obstruction

Select prostatic obstruction (N40). Chronic pyelonephritis, selected by Rule l, can be considered a direct sequel of prostatic obstruction.

Example 28:

I (a) Acute nephritis, scarlet fever

Select scarlet fever (A38). Acute nephritis, selected by Rule 2 (see Example 21), can be considered a direct sequel of scarlet fever.

Example 29:

I (a) Nephrectomy
II Clear cell carcinoma of kidney

Select clear cell carcinoma of kidney (C64). There is no doubt that the nephrectomy was performed for the malignant neoplasm of kidney.

Example 30:

I (a) Acute anaemia
(b) Haematemesis
(c) Bleeding of oesophageal varices
(d) Portal hypertension
II Cirrhosis of liver

Select cirrhosis of liver (K74.6). Portal hypertension, selected by the General Principle, can be considered a direct consequence of cirrhosis of liver.

Example 31:

I (a) Hypostatic pneumonia, cerebral
(b) Haemorrhage and cancer of breast

Select cerebral haemorrhage (I61.9). Hypostatic pneumonia, selected by Rule 2, can be considered a direct sequel of either of the other conditions reported; the one mentioned first is selected.

Example 32:

I (a) Pulmonary infarction
II Left pneumonectomy for carcinoma of lung 3 weeks ago

Select carcinoma of lung (C34.9).

4.1.8 Modification of the selected cause

The selected cause of death is not necessarily the most useful and informative condition for tabulation. For example, if senility or some generalized disease such as hypertension or atherosclerosis has been selected, this is less useful than if a manifestation or result of aging or disease had been chosen. It may sometimes be necessary to modify the selection to conform with the requirements of the classification, either for a single code for two or more causes jointly reported or for preference for a particular cause when reported with certain other conditions.

The modification rules that follow are intended to improve the usefulness and precision of mortality data and should be applied after selection of the originating antecedent cause. The interrelated processes of selection and modification have been separated for clarity.

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Some of the modification rules require further application of the selection rules, which will not be difficult for experienced coders, but it is important to go through the process of selection, modification and, if necessary, reselection.

4.1.9 The modification rules

Rule A. Senility and other ill-defined conditions

Where the selected cause is ill-defined and a condition classified elsewhere is reported on the certificate, reselect the cause of death as if the ill-defined condition had not been reported, except to take account of that condition if it modifies the coding. The following conditions are regarded as ill-defined: I46.9 (Cardiac arrest, unspecified); I95.9 (Hypotension, unspecified); I99 (Other and unspecified disorders of circulatory system); J96.0 (Acute respiratory failure); J96.9 (Respiratory failure, unspecified); P28.5 (Respiratory failure of newborn); R00-R94 or R96-R99 (Symptoms, signs and abnormal clinical and laboratory findings, not elsewhere classified). Note that R95 (Sudden infant death syndrome) is not regarded as ill-defined.

Rule B. Trivial conditions

Where the selected cause is a trivial condition unlikely to cause death and a more serious condition is reported, reselect the underlying cause as if the trivial condition had not been reported. If the death was the result of an adverse reaction to treatment of the trivial condition, select the adverse reaction.

Rule C. Linkage

Where the selected cause is linked by a provision in the classification or in the notes for use in underlying cause mortality coding with one or more of the other conditions on the certificate, code the combination.

Where the linkage provision is only for the combination of one condition specified as due to another, code the combination only when the correct causal relationship is stated or can be inferred from application of the selection rules.

Where a conflict in linkages occurs, link with the condition that would have been selected if the cause initially selected had not been reported. Make any further linkage that is applicable.

Rule D. Specificity

Where the selected cause describes a condition in general terms and a term that provides more precise information about the site or nature of this condition is reported on the certificate, prefer the more informative term. This rule will often apply when the general term becomes an adjective, qualifying the more precise term.

Rule E. Early and late stages of disease

Where the selected cause is an early stage of a disease and a more advanced stage of the same disease is reported on the certificate, code to the more advanced stage. This rule does not apply to a “chronic” form reported as due to an “acute” form unless the classification gives special instructions to that effect.

Rule F. Sequelae

Where the selected cause is an early form of a condition for which the classification provides a separate “Sequelae of ...” category, and there is evidence that death occurred from residual effects of this condition rather than from

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those of its active phase, code to the appropriate "Sequelae of ..." category.

"Sequelae of ..." categories are as follows: B90-B94, E64.-, E68, G09, I69, O97 and Y85-Y89.

4.1.10 Examples of the modification rules

Rule A. Senility and other ill-defined conditions

Where the selected cause is ill-defined and a condition classified elsewhere is reported on the certificate, reselect the cause of death as if the ill-defined condition had not been reported, except to take account of that condition if it modifies the coding. The following conditions are regarded as ill-defined: I46.9 (Cardiac arrest, unspecified); I95.9 (Hypotension, unspecified); I99 (Other and unspecified disorders of circulatory system); J96.0 (Acute respiratory failure); J96.9 (Respiratory failure, unspecified); P28.5 (Respiratory failure of newborn); R00-R94 and R96-R99 (Symptoms, signs and abnormal clinical and laboratory findings, not elsewhere classified). Note that R95 (Sudden infant death syndrome) is not regarded as ill-defined.

Example 33: I (a) Senility and hypostatic pneumonia
(b) Rheumatoid arthritis

Code to rheumatoid arthritis (M06.9). Senility, selected by Rule 2 (see Example 19), is ignored and the General Principle applied.

Example 34: I (a) Anaemia
(b) Splenomegaly

Code to splenomegalic anaemia (D64.8). Splenomegaly, selected by the General Principle, is ignored but modifies the coding.

Example 35: I (a) Myocardial degeneration and
(b) emphysema
(c) Senility

Code to myocardial degeneration (I51.5). Senility, selected by the General Principle, is ignored and Rule 2 applied.

Example 36: I (a) Cough and haematemesis

Code to haematemesis (K92.0). Cough, selected by Rule 2, is ignored.

Example 37: I (a) Terminal pneumonia
(b) Spreading gangrene and cerebrovascular
(c) infarction

Code to cerebrovascular infarction (I63.9). Gangrene, selected by Rule 1, is ignored and the General Principle is applied.

Rule B. Trivial conditions

Where the selected cause is a trivial condition unlikely to cause death and a more serious condition is reported, reselect the underlying cause as if the trivial condition had not been reported. If the death was the

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result of an adverse reaction to treatment of the trivial condition, select the adverse reaction.

Example 38: I (a) Dental caries
II Cardiac arrest

Code to cardiac arrest (I46.9). Dental caries, selected by the General Principle, is ignored.

Example 39: I (a) Ingrowing toenail and acute renal failure

Code to acute renal failure (N17.9). Ingrowing toenail, selected by Rule 2, is ignored.

Example 40: I (a) Intraoperative haemorrhage
(b) Tonsillectomy
(c) Hypertrophy of tonsils

Code to haemorrhage during surgical operation (Y60.0).

Example 41: I (a) Bursitis and ulcerative colitis

Code to ulcerative colitis (K51.9). Bursitis, selected by Rule 2 (see Example 20), is ignored.

Example 42: I (a) Paronychia
II Tetanus

Code to tetanus (A35). Paronychia, selected by the General Principle, is ignored.

Rule C. Linkage

Where the selected cause is linked by a provision in the classification or in the notes for use in underlying cause mortality coding with one or more of the other conditions on the certificate, code the combination. Where the linkage provision is only for the combination of one condition specified as due to another, code the combination only when the correct causal relationship is stated or can be inferred from application of the selection rules. Where a conflict in linkages occurs, link with the condition that would have been selected if the cause initially selected had not been reported. Make any further linkage that is applicable.

Example 43: I (a) Intestinal obstruction
(b) Femoral hernia

Code to femoral hernia with obstruction (K41.3).

Example 44: I (a) Right bundle-branch block and Chagas' disease

Code to Chagas' disease with heart involvement (B57.2). Right bundle-branch block, selected by Rule 2, links with Chagas' disease.

Example 45: I (a) Acute myocardial infarction
(b) Atherosclerotic heart disease
(c) Influenza

Code to acute myocardial infarction (I21.9). Atherosclerotic heart disease, selected by Rule 1 (see Example 13), links with acute myocardial infarction.

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Example 46: I (a) Cerebral infarction and hypostatic pneumonia
(b) Hypertension and diabetes
(c) Atherosclerosis

Code to cerebral infarction (I63.9). Atherosclerosis, selected by Rule l (see Example 15), links with hypertension, which itself links with cerebral infarction.

Example 47: I (a) Cardiac dilatation and renal sclerosis
(b) Hypertension

Code to hypertensive heart and renal disease (I13.9). All three conditions combine.

Example 48: I (a) Stroke
(b) Atherosclerosis and hypertensive heart
(c) disease

Code to hypertensive heart disease (I11.9). Atherosclerosis, selected by Rule l, links with hypertensive heart disease since hypertensive heart disease would have been selected by the General Principle if atherosclerosis had not been reported.

Example 49: I (a) Stroke and hypertensive
(b) heart disease
(c) Atherosclerosis

Code to stroke (I64). Atherosclerosis, selected by the General Principle, links with stroke since this condition would have been selected by Rule 2 if atherosclerosis had not been reported.

Example 50: I (a) Secondary polycythaemia
(b) Pulmonary emphysema
(c) Chronic bronchitis

Code to obstructive chronic bronchitis (J44.8).Chronic bronchitis, selected by the General Principle, links with emphysema.

Example 51: I (a) Cardiac dilatation
(b) Hypertension
II Atrophy of the kidneys

Code to hypertensive heart and renal disease I13.9. All three conditions combine.

Example 52: I (a) Bronchopneumonia (aspiration)
(b) Convulsions
(c) Tuberculous meningitis
II Pulmonary tuberculosis

Code to pulmonary tuberculosis (A16.2). Tuberculous meningitis, selected by the General Principle, is not to be used with mention of pulmonary tuberculosis.

Example 53: I (a) Occipital fracture

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(b) Fall following epileptic convulsions

Code to epileptic convulsions (G40.9). Fall, selected by Rule 1, links with epileptic convulsions.

Example 54: I (a) Cardiac arrest
II Chagas' disease

Code to Chagas' disease with heart involvement (B57.2). Cardiac arrest, selected by the General Principle, links with Chagas' disease.

Rule D. Specificity

Where the selected cause describes a condition in general terms and a term that provides more precise information about the site or nature of this condition is reported on the certificate, prefer the more informative term. This rule will often apply when the general term becomes an adjective, qualifying the more precise term.

Example 55: I (a) Cerebral infarction
(b) Cerebrovascular accident

Code to cerebral infarction (I63.9).

Example 56: I (a) Rheumatic heart disease, mitral stenosis

Code to rheumatic mitral stenosis (I05.0).

Example 57: I (a) Meningitis
(b) Tuberculosis

Code to tuberculous meningitis (A17.0). The conditions are stated in the correct causal relationship.

Example 58: I (a) Severe hypertension in pregnancy
II Eclamptic convulsions

Code to eclampsia in pregnancy (O15.0).

Example 59: I (a) Aneurysm of aorta
(b) Syphilis

Code to syphilitic aneurysm of aorta (A52.0). The conditions are stated in the correct causal relationship.

Example 60: I (a) Pericarditis
(b) Uraemia and pneumonia

Code to uraemic pericarditis (N18.8). Uraemia, selected by Rule l (see Example 14), modifies the pericarditis.

Rule E. Early and late stages of disease

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Where the selected cause is an early stage of a disease and a more advanced stage of the same disease is reported on the certificate, code to the more advanced stage. This rule does not apply to a "chronic" form reported as due to an "acute" form unless the classification gives special instructions to that effect.

Example 61: I (a) Tertiary syphilis
(b) Primary syphilis

Code to tertiary syphilis (A52.9).

Example 62: I (a) Eclampsia during pregnancy
(b) Pre-eclampsia

Code to eclampsia during pregnancy (O15.0).

Example 63: I (a) Chronic myocarditis
(b) Acute myocarditis

Code to acute myocarditis (I40.9).

Example 64: I (a) Chronic nephritis
(b) Acute nephritis

Code to chronic nephritis, unspecified (N03.9), as special instruction is given to this effect.

Rule F. Sequelae

Where the selected cause is an early form of a condition for which the classification provides a separate "Sequelae of ..." category, and there is evidence that death occurred from residual effects of this condition rather than from those of its active phase, code to the appropriate "Sequelae of ..." category.

"Sequelae of ..." categories are as follows: B90-B94, E64.-, E68, G09, I69, O97 and Y85-Y89.

Example 65: I (a) Pulmonary fibrosis
(b) Old pulmonary tuberculosis

Code to sequelae of respiratory tuberculosis (B90.9).

Example 66: I (a) Bronchopneumonia
(b) Curvature of spine
(c) Rickets in childhood

Code to sequelae of rickets (E64.3).

Example 67: I (a) Hydrocephalus
(b) Tuberculous meningitis

Code to sequelae of tuberculous meningitis (B90.0).

Example 68: I (a) Hypostatic pneumonia
(b) Hemiplegia
(c) Cerebrovascular accident (l0 years)

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Code to sequelae of cerebrovascular accident (I69.4).

Example 69: I (a) Chronic nephritis
(b) Scarlet fever

Code to sequelae of other specified infectious and parasitic diseases (B94.8). The description of the nephritis as chronic implies that the scarlet fever is no longer in its active phase.

4.1.11 Notes for use in underlying cause mortality coding

The following notes often indicate that if the provisionally selected code, as indicated in the left-hand column, is present with one of the conditions listed below it, the code to be used is the one shown in bold type. There are two types of combination: "with mention of" means that the other condition may appear anywhere on the certificate; "when reported as the originating antecedent cause of" means that the other condition must appear in a correct causal relationship or be otherwise indicated as being "due to" the originating antecedent cause.

A00-B99 Certain infectious and parasitic diseases

Except for human immunodeficiency virus [HIV] disease (B20-B24), *when reported as the originating antecedent cause of* a malignant neoplasm, code **C00-C97**.

A15.- Respiratory tuberculosis, bacteriologically and histologically confirmed
A16.- Respiratory tuberculosis, not confirmed bacteriologically or histologically

with mention of:

J60-J64 (Pneumoconiosis), code **J65**

A17.- Tuberculosis of nervous system
A18.- Tuberculosis of other organs

with mention of:

A15 or A16 (Respiratory tuberculosis), code **A15**, **A16**, unless reported as the originating antecedent cause of and with a specified duration exceeding that of the condition in A15.- or A16.-

A39.2 Acute meningococcaemia
A39.3 Chronic meningococcaemia
A39.4 Meningococcaemia, unspecified

with mention of:

A39.0 (Meningococcal meningitis), code **A39.0**
A39.1 (Waterhouse-Friderichsen syndrome), code **A39.1**

A40.- Streptococcal septicaemia
A41.- Other septicaemia
A46 Erysipelas

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Code to these diseases when they follow a superficial injury (any condition in S00, S10, S20, S30, S40, S50, S60, S70, S80, S90, T00, T09.0, T11.0), or first-degree burn; when they follow a more serious injury, code to the external cause of the injury.

B20-B24 Human immunodeficiency virus [HIV] disease

The subcategories at B20-B23 are the only optional four-character codes for countries using the four-character version of ICD-10. These four-character subcategories are provided for use where it is not possible or not desired to use multiple-cause coding. Conditions classifiable to two or more subcategories of the same category should be coded to the .7 subcategory of the relevant category (B20 or B21). If desired, additional codes from within the block B20-B24 may be used to specify the individual conditions listed.

B22.7 HIV disease resulting in multiple diseases classified elsewhere

This subcategory should be used when conditions classifiable to two or more categories from B20-B22 are listed on the certificate. If desired, additional codes from within the block B20-B24 may be used to specify the individual conditions listed.

B95-B97 Bacterial, viral and other infectious agents

Not to be used for underlying cause mortality coding.

D50-D89 Diseases of the blood and blood-forming organs and certain disorders involving the immune mechanism

as the cause of:

B20-B24 Human immunodeficiency virus [HIV] disease and where the certificate indicates that the HIV disease is a result of a blood transfusion given as treatment for the originating condition, code **B20-B24**

E86 Volume depletion

with mention of:

A00-A09 (Intestinal infectious diseases), code **A00-A09**

E89.- Postprocedural endocrine and metabolic disorders, not elsewhere classified

Not to be used for underlying cause mortality coding. See Operations, 4.2.6.

F01-F09 Organic, including symptomatic, mental disorders

Not to be used if the underlying physical condition is known.

F10-F19 Mental and behavioural disorders due to psychoactive substance use

Fourth characters .0 (Acute intoxication) and .5 (Psychotic disorder) *with mention of* Dependence syndrome (.2), code **F10-F19** with fourth character **.2**

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F10.- Mental and behavioural disorders due to use of alcohol

with mention of:

K70.- (Alcoholic liver disease), code **K70.-**

F10.2 Dependence syndrome due to use of alcohol

with mention of:

F10.4, F10.6, F10.7 Withdrawal state with delirium, Amnesic syndrome, Residual and late-onset psychotic disorder, code **F10.4, F10.6, F10.7**

F17.- Mental and behavioural disorders due to use of tobacco

when reported as the originating antecedent cause of:

C34.- (Malignant neoplasm of bronchus and lung), code **C34.-**
I20-I25 (Ischaemic heart disease), code **I20-I25**
J40-J47 (Chronic lower respiratory disease), code **J40-J47**

F70-F79 Mental retardation

Not to be used if the underlying physical condition is known

G25.5 Other chorea

with mention of:

I00-I02 (Acute rheumatic fever), code **I02.-**
I05-I09 (Chronic rheumatic heart disease), code **I02.-**

G81.- Hemiplegia
G82.- Paraplegia and tetraplegia
G83.- Other paralytic syndromes

Not to be used if the cause of the paralysis is known.

G97.- Postprocedural disorders of nervous system, not elsewhere classified

Not to be used for underlying cause mortality coding. See Operations, 4.2.6.

H54.- Blindness and low vision

Not to be used if the antecedent condition is known.

H59.- Postprocedural disorders of eye and adnexa, not elsewhere classified

Not to be used for underlying cause mortality coding. See Operations, 4.2.6.

H90.- Conductive and sensorineural hearing loss
H91.- Other hearing loss

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Not to be used if the antecedent condition is known.

H95.- Postprocedural disorders of ear and mastoid process, not elsewhere classified

Not to be used for underlying cause mortality coding. See Operations, 4.2.6.

I05.8 Other mitral valve diseases
I05.9 Mitral valve disease, unspecified

when of unspecified cause with mention of:

I34.- (Nonrheumatic mitral valve disorders), code **I34.-**

I09.1 Rheumatic diseases of endocardium, valve unspecified
I09.9 Rheumatic heart disease, unspecified

with mention of:

I05-I08 (Chronic rheumatic heart disease), code **I05-I08**

I10 Essential (primary) hypertension

with mention of:

I11.- (Hypertensive heart disease), code **I11.-**
I12.- (Hypertensive renal disease), code **I12.-**
I13.- (Hypertensive heart and renal disease), code **I13.-**
I20-I21 (Ischaemic heart disease), code **I20-I25**
I60-I69 (Cerebrovascular disease), code **I60-I69**
N00.- (Acute nephritic syndrome), code **N00.-**
N01.- (Rapidly progressive nephritic syndrome), code **N01.-**
N03.- (Chronic nephritic syndrome), code **N03.-**
N04.- (Nephrotic syndrome), code **N04.-**
N05.- (Unspecified nephritic syndrome), code **N05.-**
N18.- (Chronic renal failure), code **I12.-**
N19 (Unspecified renal failure), code **I12.-**
N26 (Unspecified contracted kidney), code **I12.**

when reported as the originating antecedent cause of:
H35.0 (Background retinopathy and other vascular changes), code **H35.0**
I05-I09 (Conditions classifiable to I05-I09 but not specified as rheumatic), code **I34-I38**
I34-I38 (Nonrheumatic valve disorders), code **I34-I38**
I50.- (Heart failure), code **I11.0**
I51.4- (Complications and ill-defined
I51.9 descriptions of heart disease), code **I11.-**

I11.- Hypertensive heart disease

with mention of:

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I12.- (Hypertensive renal disease), code **I13.-**
I13.- (Hypertensive heart and renal disease), code **I13.-**
I20-I25 (Ischaemic heart disease), code **I20-I25**
N18.- (Chronic renal failure), code **I13.-**
N19 (Unspecified renal failure), code **I13.-**
N26 (Unspecified contracted kidney), code **I13.-**

I12.- Hypertensive renal disease

with mention of:

I11.- (Hypertensive heart disease), code **I13.-**
I13.- (Hypertensive heart and renal disease), code **I13.-**
I20-I25 (Ischaemic heart disease), code **I20-I25**

when reported as the originating antecedent cause of:

I50.- (Heart failure), code **I13.0**
I51.4- (Complications and ill-defined
I51.9 descriptions of heart disease), code **I13.-**

I13.- Hypertensive heart and renal disease

with mention of:

I20-I25 (Ischaemic heart disease), code **I20-I25**

I15.- Secondary hypertension

Not to be used for underlying cause mortality coding. If the cause is not stated, code to Other ill-defined and unspecified causes of mortality (R99).

I20.- Angina pectoris
I24.- Other acute ischaemic heart diseases
I25.- Chronic ischaemic heart disease

with mention of:

I21.- (Acute myocardial infarction), code **I21.-**
I22.- (Subsequent myocardial infarction), code **I22.-**

I21.- Acute myocardial infarction

with mention of:

I22.- (Subsequent myocardial infarction), code **I22.-**

I23.- Certain current complications following acute myocardial infarction

Not to be used for underlying cause mortality coding. Use code **I21.-** or **I22.-** as appropriate.

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I24.0 Coronary thrombosis not resulting in myocardial infarction

Not to be used for underlying cause mortality coding. For mortality the occurrence of myocardial infarction is assumed and assignment made to **I21.-** or **I22.-** as appropriate

I27.9 Pulmonary heart disease, unspecified

with mention of:

M41.- (Scoliosis), code **I27.1**

I44.- Atrioventricular and left bundle-branch block
I45.- Other conduction disorders
I46.- Cardiac arrest
I47.- Paroxysmal tachycardia
I48 Atrial fibrillation and flutter
I49.- Other cardiac arrhythmias
I50.- Heart failure
I51.4- Complications and ill-defined descriptions of heart
I51.9 disease

with mention of:

B57.- (Chagas' disease), code **B57.-**
I20-I25 (Ischaemic heart diseases), code **I20-I25**

I50.- Heart failure
I51.9 Heart disease, unspecified

with mention of:

M41.- (Scoliosis), code **I27.1**

I50.9 Heart failure, unspecified
I51.9 Heart disease, unspecified

with mention of:

J81 (Pulmonary oedema), code **I50.1**

I65.- Occlusion and stenosis of precerebral arteries, not resulting in cerebral infarction

I66.- Occlusion and stenosis of cerebral arteries, not resulting in cerebral infarction

Not to be used for underlying cause mortality coding. For mortality, the occurrence of cerebral infarction is assumed and assignment made to **I63.-**.

I67.2 Cerebral atherosclerosis

with mention of:

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I60-I64 (Cerebral haemorrhage, cerebral infarction or stroke), code **I60-I64**

when reported as the originating antecedent cause of conditions in:

F03 (Unspecified dementia), code **F01.-**
G20 (Parkinson's disease), code **G20**

I70.- Atherosclerosis

with mention of:

I10-I13 (Hypertensive disease), code **I10-I13**
I20-I25 (Ischaemic heart diseases), code **I20-I25**
I51.4 (Myocarditis, unspecified), code **I51.4**
I51.5 (Myocardial degeneration), code **I51.5**
I51.6 (Cardiovascular disease, unspecified), code **I51.6**
I51.8 (Other ill-defined heart diseases), code **I51.8**
I51.9 (Heart disease, unspecified), code **I51.9**
I60-I69 (Cerebrovascular diseases), code **I60-I69**

when reported as the originating antecedent cause of:

I05-I09 (Conditions classifiable to I05-I09 but not specified as rheumatic), code **I34-I38**
I34-I38 (Nonrheumatic valve disorders), code **I34-I38**
I71-I78 (Other diseases of arteries, arterioles and capillaries), code **I71-I78**
K55.- (Vascular disorders of intestine), code **K55.-**
N26 (Unspecified contracted kidney), code **I12.-**

I70.9 Generalized and unspecified atherosclerosis

with mention of:

R02 (Gangrene, not elsewhere classified), code **I70.2**

when reported as the originating antecedent cause of:

F03 (Unspecified dementia), code **F01.-**
G20 (Parkinson's disease), code **G20**

I97.- Postprocedural disorders of circulatory system, not elsewhere classified

Not to be used for underlying cause mortality coding. See Operations, 4.2.6.

J00 Acute nasopharyngitis [common cold]
J06.- Acute upper respiratory infections of multiple and unspecified sites

when reported as the originating antecedent cause of:

G03.8 (Meningitis), code **G03.8**
G06.0 (Intracranial abscess and granuloma), code **G06.0**

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H65-H66 (Otitis media), code **H65-H66**
H70.- (Mastoiditis and related conditions), code **H70.-**
J10-J18 (Influenza and pneumonia), code **J10-J18**
J20-J21 (Bronchitis and bronchiolitis), code **J20-J21**
J40-J42 (Unspecified and chronic bronchitis), code **J40-J42**
J44.- (Other chronic obstructive pulmonary disease), code **J44.-**
N00.- (Acute nephritic syndrome), code **N00.-**

J20.- Acute bronchitis

with mention of:

J41.- (Simple and mucopurulent chronic bronchitis), code **J41.-**
J42 (Unspecified chronic bronchitis), code **J42**
J44 (Other chronic obstructive pulmonary disease), code **J44**

J40 Bronchitis, not specified as acute or chronic
J41.- Simple and mucopurulent chronic bronchitis
J42 Unspecified chronic bronchitis

with mention of:

J43.- (Emphysema) code **J44.-**
J44.- (Other chronic obstructive pulmonary disease) code **J44.-**

when reported as the originating antecedent cause of:

J45.- (Asthma), code **J44.-** (but see also note at J45.-, J46, below)

J43.- Emphysema

with mention of:

J40 (Bronchitis, not specified as acute or chronic), code **J44.-**
J41.- (Simple and mucopurulent chronic bronchitis), code **J44.-**
J42 (Unspecified chronic bronchitis), code **J44.-**

J45.- Asthma
J46 Status asthmaticus

When asthma and bronchitis (acute)(chronic) or other chronic obstructive pulmonary disease are reported together on the medical certificate of cause of death, the underlying cause should be selected by applying the General Principle or Rules 1, 2 or 3 in the normal way. Neither term should be treated as an adjectival modifier of the other.

J60-J64 Pneumoconiosis

with mention of:

A15-A16 (Respiratory tuberculosis), code **J65**

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J81 Pulmonary oedema

with mention of:

I50.9 (Heart failure, unspecified), code **I50.1**
I51.9 (Heart disease, unspecified), code **I50.1**

J95.- Postprocedural respiratory disorders, not elsewhere classified

Not to be used for underlying cause mortality coding. See Operations, 4.2.6.

K91.- Postprocedural disorders of digestive system, not elsewhere classified

Not to be used for underlying cause mortality coding. See Operations, 4.2.6.

M41.- Scoliosis

with mention of:

I27.9 (Pulmonary heart disease, unspecified), code **I27.1**
I50.- (Heart failure), code **I27.1**
I51.9 (Heart disease, unspecified), code **I27.1**

M96.- Postprocedural musculoskeletal disorders, not elsewhere classified

Not to be used for underlying cause mortality coding. See Operations, 4.2.6.

N00.- Acute nephritic syndrome

when reported as the originating antecedent cause of:

N03.- (Chronic nephritic syndrome), code **N03.-**

N18.- Chronic renal failure
N19 Unspecified renal failure
N26 Unspecified contracted kidney

with mention of:

I10 (Essential (primary) hypertension), code **I12.-**
I11.- (Hypertensive heart disease), code **I13.-**
I12.- (Hypertensive renal disease), code **I12.-**

N46 Male infertility
N97.- Female infertility

Not to be used if the causative condition is known.

N99.- Postprocedural disorders of genitourinary system, not elsewhere classified

Not to be used for underlying cause mortality coding. See Operations, 4.2.6.

O08.- Complications following abortion and ectopic and molar pregnancy

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Not to be used for underlying cause mortality coding. Use categories O00-O07.

O30.- Multiple gestation

Not to be used for underlying cause mortality coding if a more specific complication is reported.

O32.- Maternal care for known or suspected malpresentation of fetus

with mention of :

O33.- (Maternal care for known or suspected disproportion), code **O33.-**

O33.9 Fetopelvic disproportion

with mention of:

O33.0-O33.3 (Disproportion due to abnormality of maternal pelvis), code **O33.0-O33.3**

O64.- Obstructed labour due to malposition and malpresentation of fetus

with mention of:

O65.- (Obstructed labour due to maternal pelvic abnormality), code **O65.-**

O80-O84 Method of delivery

Not to be used for underlying cause mortality coding. If no other cause of maternal mortality is reported, code to Complication of labour and delivery, unspecified (O75.9)

P07.- Disorders related to short gestation and low birth weight, not elsewhere classified
P08.- Disorders related to long gestation and high birth weight

Not to be used if any other cause of perinatal mortality is reported.

R69.- Unknown and unspecified causes of morbidity

Not to be used for underlying cause mortality coding. Use R95-R99 as appropriate.

S00-T98 Injury, poisoning and certain other consequences of external causes

Not to be used for underlying cause mortality coding except as an additional code to the relevant category in V01-Y89

S02.- Fracture of skull and facial bones

When more than one site is mentioned, code to multiple fractures involving skull and facial bones, **S02.7**

S06.- Intracranial injury

When a fracture of the skull or facial bones is associated with an intracranial injury, priority should be given to the fracture.

with mention of:

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S02.- (Fracture of skull or facial bones), code **S02.-**

T36-T50 Poisoning by drugs, medicaments and biological substances (accidental poisoning and poisoning of undetermined intent by alcohol or dependence-producing drugs)

with mention of:

F10-F19 with fourth character .2 (alcohol dependence or drug dependence), code **F10-F19** with fourth character **.2**

T79.- Certain early complications of trauma, not elsewhere classified

Not to be used if the nature of the antecedent injury is known.

V01-X59 Accidents

with mention of:

A35 (Tetanus), code **A35**

resulting from:

G40-G41 (Epilepsy), code **G40-G41**

X40-X49 Accidental poisoning by and exposure to noxious substances

Y10-Y15 Poisoning by and exposure to noxious substances, undetermined intent (poisoning by alcohol or dependence-producing drugs)

with mention of:

F10-F19 with fourth character .2 (alcohol dependence or drug dependence) code, **F10-F19** with fourth character **.2**

Y90-Y98 Supplementary factors related to causes of morbidity and mortality classified elsewhere

Not to be used for underlying cause mortality coding.

Z00-Z99 Factors influencing health status and contact with health services

Not to be used for underlying cause mortality coding.

4.1.12 Summary of linkages by code number

When the selected cause is listed in the first column of Table 1, and one or more of the causes listed in the second column have been entered anywhere on the certificate, code as indicated in the fourth column.

When the selected cause is listed in the first column and appears on the certificate as a cause of one of the diseases listed in the third column, code as indicated in the fourth column.

Table 1. Summary of linkages by code number

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Selected cause	With mention of:	As cause of:	Resulting linked code
A00-B19 }			
B25-B99 }		C00-C97	C00-C97
A15.-, A16.-	J60-J64		J65
A17.-, A18.-	A15.-, A16.-		A15.-, A16.-
A39.2-A39.4	A39.0, A39.1		A39.0, A39.1
D50-D59	B20-B24		B20-B24
E86	A00-A09		A00-A09
F10-F19 **(F1x.0)}**	F10-F19 **(F1x.2)**		F10-F19 **(F1x.2) (F1x.5) }**
F10	K70.-		K70.-
F10.2	F10.4, F10.6, F10.7		F10.4, F10.6, F10.7
F17.-		C34.-	C34.-
		I20-I25	I20-I25
		J40-J47	J40-J47
G25.5	I00-I02		I02.-
	I05-I09		I02.-
I05.8 **}**			
I05.9 **}**			
(of unspecified }			
cause) }	I34.-		I34.-
I09.1 **}**			
I09.9 **}**	I05-I08		I05-I08
I10	I11.-		I11.-
	I12.-		I12.-
	I13.-		I13.-
	I20-I25		I20-I25
	I60-I69		I60-I69
	N00.-		N00.-
	N01.-		N01.-
	N03-N05		N03-N05
	N18.-		I12.-
	N19		I12.-
	N26		I12.-
		H35.0	H35.0
		I05-I09 **(not specified as rheumatic)**	I34-I38
		I34-I38	I34-I38
		I50.-	I11.0
		I51.4-I51.9	I11.-

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I11.-	I12.-		I13.-
	I13.-		I13.-
	I20-I25		I20-I25
	N18.-		I13.-
	N19		I13.-
	N26		I13.-
I12.-	I11.-		I13.-
	I13.-		I13.-
	I20-I25		I20-I25
		I50.-	I13.0
		I51.4 -I51.9	I13.-
I13.-	I20-I25		I20-I25
I20.- **}**			
I24.- **}**	I21.-		I21.-
I25.- **}**	I22.-		I22.-
I21.-	I22.-		I22.-
I27.9	M41.-		I27.1
I44-I50 **}**	B57.-		B57.-
I51.4-I51.9 **}**	I20-I25		I20-I25
I50.- **}**			
I51.9 **}**	M41.-		I27.1
I50.9 **}**			
I51.9 **}**	J81		I50.1
I67.2	I60-I64		I60-I64
		F03	F01.-
		G20	G20
I70.-	I10-I13		I10-I13
	I20-I25		I20-I25
	I51.4		I51.4
	I51.5		I51.5
	I51.6		I51.6
	I51.8		I51.8
	I51.9		I51.9
	I60-I69		I60-I69
		I05-I09 **(not speficied as rheumatic)**	I34-I38
		I34-I38	I34-I38
		I71-I78	I71-I78
		K55.-	K55.-
		N26	I12.-
I70.9	R02		I70.2

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		F03	F01.-
		G20	G20
J00 **}**			
J06.- **}**		G03.8	G03.8
		G06.0	G06.0
		H65-H66	H65-H66
		H70.-	H70.-
		J10-J18	J10-J18
		J20-J21	J20-J21
		J40	J40-J42
		J44.-	J44.-
		N00.-	N00.-
J20.-	J41.-		J41.-
	J42		J42
	J44.-		J44.-
J40 **}**			
J41.- **}**	J43.-		J44.-
J42 **}**	J44.-		J44.-
		J45.-	J44.-
J43.-	J40		J44.-
	J41.-		J44.-
	J42		J44.-
J60-J64	A15.-		J65
	A16.-		J65
J81	I50.9		I50.1
	I51.9		I50.1
M41.-	I27.9		I27.1
	I50.-		I27.1
	I51.9		I27.1
N00.-		N03.-	N03.-
N18.- **}**			
N19 **}**			
N26 **}**	I10		I12.-
	I11.-		I13.-
	I12.-		I12.-
O32.-	O33.-		O33.-
O33.9	O33.0-O33.3		O33.0-O33.3
O64.-	O65.-		O65.-
S06.-	S02.-		S02.-
T36-T50	F10-F19 **(F1x.2)**		F10-F19 **(F1x.2)**
V01-X59	A35		A35
X40-X49 **}**			

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Y10-Y15 } F10-F19 **(F1x.2)** F10-F19 **(F1x.2)**

Table 2. Summary of codes not to be used in underlying cause mortality coding[a]

Codes not to be used for underlying cause mortality coding (code to item in parentheses; if no code is indicated, code to R99)		Not to be used if the underlying cause is known
B95-B97		F01-F09
E89.-		F70-F79
G97.-		G81.-
H59.-		G82.-
H95.-		G83.-
I15.-		H54.-
I23.-	**(code to** I21 **or** I22**)**	H90-H91
I24.0	**(code to** I21 **or** I22**)**	N46
I65.-	**(code to** I63**)**	N97.-
I66.-	**(code to** I63**)**	O30.-
I97.-		P07.-
J95.-		P08.-
K91.-		T79.-
M96.-		
N99.-		
O08.-		
O80-O84	**(code to** O75.9**)**	
R69.-	**(code to** R95-R99**)**	
S00-T98	**(code to** V01-Y89**)**	
Y90-Y98		
Z00-Z99		

[a] In addition to asterisk codes (see section 3.1.3)

4.2 Notes for interpretation of entries of causes of death

The foregoing rules will usually determine the underlying cause of death to be used for primary mortality tabulation. Each country will need to amplify the rules, depending upon the completeness and quality of medical certification. The information in this section will help in formulating such additional instructions.

4.2.1 Assumption of intervening cause

Frequently on the medical certificate, one condition is indicated as due to another, but the first one is not a direct consequence of the second one. For example, haematemesis may be stated as due to cirrhosis of the liver, instead of being reported as the final event of the sequence, liver cirrhosis→portal hypertension→ruptured oesophageal varices→haematemesis.

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The assumption of an intervening cause in Part I is permissible in accepting a sequence as reported, but it must not be used to modify the coding.

Example 1: I (a) Cerebral haemorrhage
(b) Chronic nephritis

Code to chronic nephritis (N03.9). It is necessary to assume hypertension as a condition intervening between cerebral haemorrhage and the underlying cause, chronic nephritis.

Example 2: I (a) Mental retardation
(b) Premature separation of placenta

Code to premature separation of placenta affecting fetus or newborn (P02.1). It is necessary to assume birth trauma, anoxia or hypoxia as a condition intervening between mental retardation and the underlying cause, premature separation of placenta.

4.2.2 Interpretation of "highly improbable"

The expression "highly improbable" has been used since the Sixth Revision of the ICD to indicate an unacceptable causal relationship. As a guide to the acceptability of sequences in the application of the General Principle and the selection rules, the following relationships should be regarded as "highly improbable":

(a) any infectious disease may be accepted as "due to" disorders of the immune mechanism such as human immunodeficiency virus [HIV] disease or AIDS;

(b) an infectious or parasitic disease (A00-B99) reported as "due to" any disease outside this chapter, except that:

- diarrhoea and gastroenteritis of presumed infectious origin (A09))
- septicaemia (A40-A41)) may be accepted
- erysipelas (A46)) as "due to" any
- gas gangrene (A48.0)) disease
- Vincent's angina (A69.1))
- mycoses (B35-B49))
- any infectious disease may be accepted as "due to" immunosuppression by chemicals (chemotherapy) and radiation. Any infectious disease classified to A00-B19 or B25-B64 reported as "due to" a malignant neoplasm will also be an acceptable sequence.
- varicella and zoster infections (B01-B02) may be accepted as "due to" diabetes, tuberculosis and lymphoproliferative neoplasms;

(c) a malignant neoplasm reported as "due to" any other disease, except human immunodeficiency virus [HIV] disease;

(d) haemophilia (D66, D67, D68.0-D68.2) reported as "due to" any other disease;

(e) diabetes (E10-E14) reported as "due to" any other disease except:

- haemochromatosis (E83.1),
- diseases of pancreas (K85-K86),
- pancreatic neoplasms (C25.-, D13.6, D13.7, D37.7),
- malnutrition (E40-E46);

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(f) rheumatic fever (I00-I02) or rheumatic heart disease (I05-I09) reported as "due to" any disease other than scarlet fever (A38), streptococcal septicaemia (A40), streptococcal sore throat (J02.0) and acute tonsillitis (J03.-);

(g) any hypertensive condition reported as "due to" any neoplasm except:
- endocrine neoplasms,
- renal neoplasms,
- carcinoid tumours;

(h) chronic ischaemic heart disease (I20, I25) reported as "due to" any neoplasm;

(i) any cerebrovascular disease (I60-I69) reported as "due to" a disease of the digestive system (K00-K92) or endocarditis (I05-I08, I09.1, I33-I38), except for cerebral embolism in I65-I66 or intracranial haemorrhage (I60-I62);

(j) any condition described as arteriosclerotic [atherosclerotic] reported as "due to" any neoplasm;

(k) influenza (J10-J11) reported as "due to" any other disease;

(l) a congenital anomaly (Q00-Q99) reported as "due to" any other disease of the individual, including immaturity;

(m) a condition of stated date of onset "X" reported as "due to" a condition of stated date of onset "Y", when "X" predates "Y" (but see also Example 5 in section 4.1.6);

(n) accidents (V01-X59) reported as "due to" any other cause outside this chapter except:
- (1) any accident (V01-X59) reported as "due to" epilepsy (G40-G41),
- (2) a fall (W00-W19) "due to" a disorder of bone density (M80-M85),
- (3) a fall (W00-W19) "due to" a (pathological) fracture caused by a disorder of bone density,
- (4) asphyxia reported as "due to" aspiration of mucus, blood (W80) or vomitus (W78) as a result of diesease conditions,
- (5) aspiration of food (liquid or solid) of any kind (W79) reported as "due to" a disease which affects the ability to swallow;

(o) suicide (X60-X84) reported as "due to" any other cause.

The above list does not cover all "highly improbable" sequences, but in other cases the General Principle should be followed unless otherwise indicated.

Acute or terminal circulatory diseases reported as due to malignant neoplasm, diabetes or asthma should be accepted as possible sequences in Part I of the certificate. The following conditions are regarded as acute or terminal circulatory diseases:

I21-I22	Acute myocardial infarction
I24.-	Other acute ischaemic heart diseases
I26.-	Pulmonary embolism
I30.-	Acute pericarditis
I33.-	Acute and subacute endocarditis
I40.-	Atrioventricular and left bundle-branch block
I45.-	Other conduction disorders
I46.-	Cardiac arrest
I47.-	Paroxysmal tachycardia

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I48	Atrial fibrillation and flutter
I49.-	Other cardiac arrhythmias
I50.-	Heart failure
I51.8	Other ill-defined heart diseases
I60-I68	Cerebrovascular diseases except I67.0-I67.5 and I67.9

4.2.3 Effect of duration on classification

In evaluating the reported sequence of the direct and antecedent causes, the interval between the onset of the disease or condition and time of death must be considered. This would apply in the interpretation of "highly improbable" relationships (see above) and in Modification Rule F (sequelae).

Categories O95 (Obstetric death of unspecified cause), O96 (Death from any obstetric cause occurring more than 42 days but less than one year after delivery) and O97 (Death from sequelae of direct obstetric causes) classify obstetric deaths according to the time elapsed between the obstetric event and the death of the woman. Category O95 is to be used when a woman dies during pregnancy, labour, delivery, or the puerperium and the only information provided is "maternal" or "obstetric" death. If the obstetric cause of death is specified, code to the appropriate category. Category O96 is used to classify deaths from direct or indirect obstetric causes that occur more than 42 days but less than a year after termination of the pregnancy. Category O97 is used to classify deaths from any direct obstetric cause which occur one year or more after termination of the pregnancy.

Conditions classified as congenital malformations, deformations and chromosomal abnormalities (Q00-Q99), even when not specified as congenital on the medical certificate, should be coded as such if the interval between onset and death and the age of the decedent indicate that the condition existed from birth.

The classification has specific categories for indicating certain diseases and injuries as the cause of sequelae or late effects. In many cases, these sequelae include conditions present one year or more after the onset of the disease or injury (see also Sequelae below).

4.2.4 Sequelae

Certain categories (B90-B94, E64.-, E68, G09, I69.-, O97 and Y85-Y89) are to be used for underlying cause mortality coding to indicate that death resulted from the late (residual) effects of a given disease or injury rather than during the active phase. Modification Rule F applies in such circumstances. Conditions reported as sequelae or residual effects of a given disease or injury should be classified to the appropriate sequela category, irrespective of the interval between the onset of the disease or injury and death. For certain conditions, deaths occurring one year or more after the onset of the disease or injury are assumed to be due to a sequela or residual effect of the condition, even though no sequela is explicitly mentioned. Guidance in interpreting sequelae is given under most of the "Sequelae of ..." categories in the tabular list.

B90.- Sequelae of tuberculosis

The sequelae include conditions specified as such or as late effects of past tuberculous disease, and residuals of tuberculosis specified as arrested, cured, healed, inactive, old, or quiescent, unless there is evidence of active tuberculosis.

B94.0 Sequelae of trachoma

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The sequelae include residuals of trachoma specified as healed or inactive and certain specified sequelae such as blindness, cicatricial entropion and conjunctival scars, unless there is evidence of active infection.

B94.1 Sequelae of viral encephalitis

The sequelae include conditions specified as such, or as late effects, and those present one year or more after onset of the causal condition.

B94.8 Sequelae of other infectious and parasitic diseases

The sequelae include conditions specified as such or as late effects and residuals of these diseases described as arrested, cured, healed, inactive, old or quiescent, unless there is evidence of active disease. Sequelae also include chronic conditions reported as due to, or residual conditions present one year or more after onset of, conditions classifiable to categories A00-B89.

E64.3 Sequelae of rickets

The sequelae include any condition specified as rachitic or due to rickets and present one year or more after onset, or stated to be a sequela or late effect of rickets.

G09 Sequelae of inflammatory diseases of central nervous system

This category is provided for the coding of sequelae of conditions classifiable to G00.-, G03-G04, G06.- and G08. Sequelae of inflammatory diseases of the central nervous system subject to dual classification (G01*-G02*, G05.-* and G07*) should be coded to the categories designated for sequelae of the underlying condition (e.g. B90.0 Sequelae of central nervous system tuberculosis). If there is no sequelae category for the underlying condition, code to the underlying condition itself.

4.2.5 Consistency between sex of patient and diagnosis

Certain categories are limited to one sex (see section 3.1.5). If, after verification, the sex and cause of death on the certificate are not consistent, the death should be coded to “Other ill-defined and unspecified causes of mortality” (R99).

4.2.6 Operations

If an operation appears on the certificate as the cause of death without mention of the condition for which it was performed or of the findings at operation, and the alphabetical index does not provide a specific code for the operation, code to the residual category for the organ or site indicated by the name of the operation (e.g. code “nephrectomy” to N28.9). If the operation does not indicate an organ or site, e.g. “laparotomy”, code to “Other illdefined and unspecified causes of mortality” (R99), unless there is a mention of a therapeutic misadventure classifiable to Y60-Y84 or a postoperative complication.

4.2.7 Malignant neoplasms

When a malignant neoplasm is considered to be the underlying cause of death, it is most important to determine the primary site. Morphology and behaviour should also be taken into consideration. Cancer is a generic term and may be used for any morphological group, although it is rarely applied to malignant neoplasms of lymphatic, haematopoietic and related tissues. Carcinoma is sometimes used incorrectly as a synonym for cancer. Some death

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certificates may be ambiguous if there was doubt about the site of the primary or imprecision in drafting the certificate. In these circumstances, if possible, the certifier should be asked to give clarification.

Failing this, the guidelines given below should be observed. The morphological types of tumours classified in Volume 1 can be found in the Alphabetical Index with their morphology code and with an indication as to the coding by site.

A. Implication of malignancy

Mention on the certificate that a neoplasm has produced metastases (secondaries) means that it must be coded as malignant, even though this neoplasm without mention of metastases would be classified to some other section of Chapter II.

Example 1: I (a) Metastatic involvement of lymph nodes
(b) Carcinoma in situ of breast

Code to malignant neoplasm of breast (C50.9).

B. Sites with prefixes or imprecise definitions

Neoplasms of sites prefixed by "peri", "para", "pre", "supra", "infra", etc. or described as in the "area" or "region" of a site, unless these terms are specifically indexed, should be coded as follows: for morphological types classifiable to one of the categories C40, C41 (bone and articular cartilage), C43 (malignant melanoma of skin), C44 (other malignant neoplasms of skin), C45 (mesothelioma), C47 (peripheral nerves and autonomic nervous system), C49 (connective and soft tissue), C70 (meninges), C71 (brain) and C72 (other parts of central nervous system), code to the appropriate subdivision of that category; otherwise code to the appropriate subdivision of C76 (other and ill-defined sites).

Example 2: I (a) Fibrosarcoma in the region of the leg

Code to malignant neoplasm of connective and soft tissue of lower limb (C49.2).

C. Malignant neoplasms of unspecified site with other reported conditions

When the site of a primary malignant neoplasm is not specified, no assumption of the site should be made from the location of other reported conditions such as perforation, obstruction, or haemorrhage. These conditions may arise in sites unrelated to the neoplasm, e.g. intestinal obstruction may be caused by the spread of an ovarian malignancy.

Example 3: I (a) Obstruction of intestine
(b) Carcinoma

Code to malignant neoplasm without specification of site (C80).

D. Malignant neoplasms with primary site indicated

If a particular site is indicated as primary, it should be selected, regardless of the position on the certificate or whether in Part I or Part II. If the primary site is stated to be unknown, see E below. The primary site may be indicated in one of the following ways:

(a) The specification of one site as primary in either Part I or II.

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Example 4: I (a) Carcinoma of bladder
II Primary in kidney

Code to malignant neoplasm of kidney (C64).

(b) The specification of other sites as "secondary", "metastases", "spread" or "carcinomatosis".

Example 5: I (a) Carcinoma of breast
(b) Secondaries in brain

Code to malignant neoplasm of breast (C50.9), since Rule 2 applies

(c) Morphology indicates a primary malignant neoplasm.

If a morphological type implies a primary site, such as hepatoma, consider this as if the word "primary" had been included.

Example 6: I (a) Metastatic carcinoma
(b) Pseudomucinous adenocarcinoma

Code to malignant neoplasm of ovary (C56), since pseudomucinous adenocarcinoma of unspecified site is assigned to the ovary in the Alphabetical Index.

If two or more primary sites or morphologies are indicated, these should be coded according to sections F, G and H, below.

E. Primary site unknown

If the statement, "primary site unknown", or its equivalent, appears anywhere on a certificate, code to the category for unspecified site for the morphological type involved (e.g. adenocarcinoma C80, fibrosarcoma C49.9, osteosarcoma C41.9), regardless of the site(s) mentioned elsewhere on the certificate.

Example 7: I (a) Secondary carcinoma of liver
(b) Primary site unknown
(c) ? Stomach ? Colon

Code to carcinoma without specification of site (C80).

Example 8: I (a) Generalized metastases
(b) Melanoma of back
(c) Primary site unknown

Code to malignant melanoma of unspecified site (C43.9).

F. Independent (primary) multiple sites (C97)

The presence of more than one primary neoplasm could be indicated by mention of two different anatomical sites or two distinct morphological types (e.g. hypernephroma and intraductal carcinoma), or by a mix of a morphological type that implies a specific site, plus a second site. It is highly improbable that one primary would be due to another primary malignant neoplasm except for the group of malignant neoplasms of lymphoid, haematopoietic and related tissue (C81-C96), within which one form of malignancy may terminate in another (e.g. leukaemia may follow non-

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Hodgkin's lymphoma).

If two or more sites mentioned in Part I are in the same organ system, see section H. If the sites are not in the same organ system and there is no indication that any is primary or secondary, code to malignant neoplasms of independent (primary) multiple sites (C97), unless all are classifiable to C81- C96, or one of the sites mentioned is a common site of metastases or the lung (see G below).

Example 9:

I (a) Cancer of stomach
(b) Cancer of breast

Code to malignant neoplasms of independent (primary) multiple sites (C97), since two different anatomical sites are mentioned and it is unlikely that one primary malignant neoplasm would be due to another.

Example 10:

I (a) Hodgkin's disease
(b) Carcinoma of bladder

Code to malignant neoplasms of independent (primary) multiple sites (C97), since two distinct morphological types are mentioned.

Example 11:

I (a) Acute lymphocytic leukaemia
(b) Non-Hodgkin's lymphoma

Code to non-Hodgkin's lymphoma (C85.9), since both are classifiable to C81-C96 and the sequence is acceptable.

Example 12:

I (a) Leukaemia
(b) Non-Hodgkin's lymphoma
(c) Carcinoma of ovary

Code to malignant neoplasms of independent (primary) multiple sites (C97), since, although two of the neoplasms are classifiable to C81-C96, there is mention of a site elsewhere.

Example 13:

I (a) Leukaemia
II Carcinoma of breast

Code to leukaemia (C95.9) because the carcinoma of breast is in Part II. When dealing with multiple sites, only sites in Part I of the certificate should be considered (see H).

G. Metastatic neoplasms

When a malignant neoplasm spreads or metastasizes it generally retains the same morphology even though it may become less differentiated. Some metastases have such a characteristic microscopic appearance that the pathologist can infer the primary site with confidence, e.g. thyroid. Widespread metastasis of a carcinoma is often called carcinomatosis.If an unqualified nonspecific term such as carcinoma or sarcoma appears with a term describing a more specific histology of the same broad group, code to the site of the more specific morphology, assuming the other to be metastatic.

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Although malignant cells can metastasize anywhere in the body, certain sites are more common than others and must be treated differently (see below). However, if one of these sites appears alone on a death certificate and is not qualified by the word "metastatic", it should be considered primary.

Common sites of metastases

Bone	Mediastinum
Brain	Meninges
Diaphragm	Peritoneum
Heart	Pleura
Liver	Retroperitoneum
Lung	Spinal cord
Lymph nodes	
Ill-defined sites (sites classifiable to C76)	

- The lung poses special problems in that it is a common site for both metastases and primary malignant neoplasms. Lung should be considered as a common site of metastases whenever it appears with sites not on this list. However, when the bronchus or bronchogenic cancer is mentioned this neoplasm should be considered primary. If lung is mentioned and the only other sites are on the list of common sites of metastases, consider lung primary.
- Malignant neoplasm of lymph nodes not specified as primary should be assumed to be secondary.

Example 14: I (a) Cancer of brain

Code to malignant neoplasm of brain (C71.9).

Example 15: I (a) Cancer of bone
(b) Metastatic carcinoma of lung

Code to malignant neoplasm of lung (C34.9), since bone is on the list of common sites of metastases and lung can therefore be assumed to be primary.

The adjective "metastatic" is used in two ways - sometimes meaning a secondary from a primary elsewhere and sometimes denoting a primary that has given rise to metastases. In order to avoid confusion, the following guidelines are proposed:

(a) Malignant neoplasm described as "metastatic from" a specified site should be interpreted as primary of that site.

Example 16: I (a) Metastatic teratoma from ovary

Code to malignant neoplasm of ovary (C56).

(b) Malignant neoplasm described as "metastatic to" a site should be interpreted as secondary of that site unless the morphology indicates a specific primary site.

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Example 17: I (a) Metastatic carcinoma to the rectum

Code to secondary malignant neoplasm of rectum (C78.5). The word "to" clearly indicates rectum as secondary.

Example 18: I (a) Metastatic osteosarcoma to brain

Code to malignant neoplasm of bone (C41.9), since this is the unspecified site of osteosarcoma.

(c) A single malignant neoplasm described as "metastatic (of)".

The terms "metastatic" and "metastatic of" should be interpreted as follows:

(i) If one site is mentioned and this is qualified as metastatic, code to malignant primary of that particular site if no morphological type is mentioned and it is not a common metastatic site (see list of common sites of metastases given above).

Example 19: I (a) Cervical cancer, metastatic

Code to malignant neoplasm of cervix (C53.9).

(ii) If no site is reported but the morphological type is qualified as metastatic, code as for primary site unspecified of the particular morphological type involved.

Example 20: I (a) Metastatic oat cell carcinoma

Code to malignant neoplasm of lung (C34.9).

(iii) If a single morphological type and a site, other than a common metastatic site (see list given above), are mentioned as metastatic, code to the specific category for the morphological type and site involved.

Example 21: I (a) Metastatic melanoma of arm

Code to malignant melanoma of skin of arm (C43.6), since in this case the ill-defined site of arm is a specific site for melanoma, not a common site of metastases classifiable to C76.

(iv) If a single morphological type is mentioned as metastatic and the site mentioned is one of the common sites of metastases except lung, code to "unspecified site" for the morphological type, unless the unspecified site is classified to C80 (malignant neoplasm without specification of site), in which case code to secondary malignant neoplasm of the site mentioned.

Example 22: I (a) Metastatic osteosarcoma of brain

Code to malignant neoplasm of bone, unspecified (C41.9), since brain is on the list of common sites of metastases.

(v) If one of the common sites of metastases, except lung, is described as metastatic and no other site or morphology is mentioned, code to secondary neoplasm of the site (C77-C79).

Example 23: I (a) Metastatic brain cancer

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Code to secondary malignant neoplasm of brain (C79.3).

Example 24: I (a) Metastatic carcinoma of lung

Code to malignant neoplasm of lung (C34.9).

(d) More than one malignant neoplasm qualified as metastatic.

(i) If two or more sites with the same morphology, not on the list of common sites of metastases, are reported and all are qualified as "metastatic", code as for primary site unspecified of the anatomical system and of the morphological type involved.

Example 25: I (a) Metastatic carcinoma of prostate
(b) Metastatic carcinoma of skin

Code to malignant neoplasm without specification of site (C80), since metastatic carcinoma of prostate is not likely to be due to metastatic carcinoma of skin; both are probably due to spread from a malignant neoplasm of unknown primary site, which should have been entered on line (c).

Example 26: I (a) Metastatic carcinoma of stomach
(b) Metastatic carcinoma of breast
(c) Metastatic carcinoma of lung

Code to malignant neoplasm without specification of site (C80), since breast and stomach do not belong to the same anatomical system and lung is on the list of common sites of metastases.

(ii) If two or more morphological types of different histological groups are qualified as metastatic, code to malignant neoplasms of independent (primary) multiple sites (C97) (see F).

Example 27: I (a) Bowel obstruction
(b) Metastatic adenocarcinoma of bowel
(c) Metastatic sarcoma of uterus

Code to malignant neoplasms of independent (primary) multiple sites (C97).

(iii) If a morphology implying site and an independent anatomical site are both qualified as metastatic, code to malignant neoplasm without specification of site (C80).

Example 28: I (a) Metastatic colonic and renal cell carcinoma

Code to malignant neoplasm without specification of site (C80).

(iv) If more than one site with the same morphology is mentioned and all but one are qualified as metastatic or appear on the list of common sites of metastases, code to the site that is not qualified as metastatic, irrespective of the order of entry or whether it is in Part I or Part II. If all sites are qualified as metastatic or on the list of common sites of metastases, including lung, code to malignant neoplasm without specification of site (C80).

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Example 29: I (a) Metastatic carcinoma of stomach
(b) Carcinoma of gallbladder
(c) Metastatic carcinoma of colon

Code to malignant neoplasm of gallbladder (C23).

Example 30: I (a) Metastatic carcinoma of ovary
(b) Carcinoma of lung
(c) Metastatic cervical carcinoma

Code to malignant neoplasm without specification of site (C80).

Example 31: I (a) Metastatic carcinoma of stomach
(b) Metastatic carcinoma of lung II Carcinoma of colon

Code to malignant neoplasm of colon (C18.9), since this is the only diagnosis not qualified as metastatic, even though it is in Part II.

(v) If all sites mentioned are on the list of common sites of metastases, code to unknown primary site of the morphological type involved, unless lung is mentioned, in which case code to malignant neoplasm of lung (C34.-).

Example 32: I (a) Cancer of liver
(b) Cancer of abdomen

Code to malignant neoplasm without specification of site (C80), since both are on the list of common sites of metastases. (Abdomen is one of the ill-defined sites included in C76.-.)

Example 33: I (a) Cancer of brain
(b) Cancer of lung

Code to cancer of lung (C34.9), since lung in this case is considered to be primary, because brain, the only other site mentioned, is on the list of common sites of metastases. (vi) If only one of the sites mentioned is on the list of common sites of metastases or lung, code to the site not on the list.

Example 34: I (a) Cancer of lung
(b) Cancer of breast

Code to malignant neoplasm of breast (C50.9), since lung in this case is considered to be a metastatic site, because breast is not on the list of common sites of metastases.

(vii) If one or more of the sites mentioned is a common site of metastases (see list given above) but two or more sites or different morphological types are also mentioned, code to malignant neoplasms of independent (primary) multiple sites (C97) (see F above).

Example 35: I (a) Cancer of liver
(b) Cancer of bladder

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(c) Cancer of colon

Code to malignant neoplasms of independent (primary) multiple sites (C97), since liver is on the list of common sites of metastases and there are still two other independent sites. (viii) If there is a mixture of several sites qualified as metastatic and several other sites are mentioned, refer to the rules for multiple sites (see F above and H below).

H. Multiple sites

When dealing with multiple sites, only sites in Part I of the certificate should be considered.

If malignant neoplasms of more than one site are entered on the certificate, the site listed as primary or not indicated whether primary or secondary should be selected (see D, E and F above).

Multiple sites with none specified as primary

(a) Notwithstanding the provisions of Rule H to consider only sites in Part I, if one of the common sites of metastases, excluding lung, and another site or morphological type are mentioned anywhere on the certificate, code to the other site. If, however, a malignant neoplasm of lymphatic, haematopoietic, or related tissue appears in Part II, only Part I should be considered.

Example 36: I (a) Cancer of stomach
(b) Cancer of liver

Code to malignant neoplasm of stomach (C16.9). Although the sequence suggests that the liver was the primary site, metastasis from liver - a common site of metastases - to stomach is improbable and it is assumed that the stomach cancer metastasized to the liver.

Example 37: I (a) Peritoneal cancer
II Mammary carcinoma

Code to malignant neoplasm of breast (C50.9), since the peritoneal cancer is presumed secondary because it is on the list of common sites of metastases.

(b) Malignant neoplasms described as one site "or" another, or if "or" is implied, should be coded to the category that embraces both sites. If no appropriate category exists, code to the unspecified site of the morphological type involved. This rule applies to all sites whether they are on the list of common sites of metastases or not.

Example 38: I (a) Carcinoma of ascending or descending colon

Code to malignant neoplasm of colon, unspecified (C18.9).

Example 39: I (a) Osteosarcoma of lumbar vertebrae or sacrum

Code to malignant neoplasm of bone, unspecified (C41.9).

(c) If two or more morphological types of malignant neoplasm occur in lymphoid, haematopoietic or related tissue (C81-C96), code according to the sequence given since these neoplasms sometimes terminate as another entity within C81-C96. Acute exacerbation of, or blastic crisis in, chronic leukaemia should be coded to the chronic form.

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Example 40: I (a) Acute lymphocytic leukaemia
(b) Non-Hodgkin's lymphoma

Code to non-Hodgkin's lymphoma (C85.9).

Example 41: I (a) Acute and chronic lymphocytic leukaemia

Code to chronic lymphocytic leukaemia (C91.1).

Multiple sites in the same organ system

If the sites mentioned are in the same organ system and are contiguous, the .8 subcategories, including those listed in Volume 1, should be used. This applies when the certificate describes the sites as one site "and" another or if the sites are mentioned on separate lines. Code to the .8 subcategory that embraces both sites. If there is any doubt about the contiguity of the sites mentioned, code to the unspecified site of the organ mentioned.

(a) If there is mention of two contiguous subsites in the same site, code to the .8 subcategory of that three-character category.

Example 42: I (a) Carcinoma of descending colon and sigmoid

Code to overlapping malignant neoplasm of colon (C18.8).

(b) If the subsites are not contiguous, code to the .9 subcategory of that three-character category.

Example 43: I (a) Carcinoma of head of pancreas
(b) Carcinoma of tail of pancreas

Code to malignant neoplasm of pancreas, unspecified (C25.9).

(c) If there is mention of two contiguous sites classified to separate three-character categories within the same body system, code to the .8 subcategory of that general body system (see list in Note 5 in the introduction to Chapter II of Volume 1).

Example 44: I (a) Carcinoma of vagina and cervix

Code to malignant neoplasm of overlapping sites of female genital organs (C57.8).

(d) If two sites are mentioned on the certificate and both are in the same organ system and have the same morphological type, code to the .9 subcategory of that organ system, as in the following list:

C26.9	Ill-defined sites within the digestive system
C39.9	Ill-defined sites within the respiratory system
C41.9	Bone and articular cartilage, unspecified
C49.9	Connective and soft tissue, unspecified
C57.9	Female genital organ, unspecifie
C63.9	Male genital organ, unspecified
C68.9	Urinary organ, unspecified
C72.9	Central nervous system, unspecified

Example 45: I (a) Pulmonary embolism

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(b) Cancer of stomach
(c) Cancer of gallbladder

Code to ill-defined sites within the digestive system (C26.9).

(e) If there is no available .8 or .9 subcategory, code to malignant neoplasms of independent (primary) multiple sites (C97).

Example 46: I (a) Cardiac arrest
(b) Carcinoma of prostate and bladder

Code to malignant neoplasms of independent (primary) multiple sites (C97), since there is no available .8 subcategory.

I. Infectious diseases and malignant neoplasms

(a) Owing to the effect of chemotherapy on the immune system, some cancer patients become prone to infectious diseases and die of them. Therefore, any infectious disease classified to A00-B19 or B25-B64 reported as "due to" cancer will be an acceptable sequence whether in Part I or II.

Example 47: I (a) Zoster
(b) Chronic lymphocytic leukaemia

Code to chronic lymphocytic leukaemia (C91.1).

(b) Except for human immunodeficiency virus [HIV] disease, no infectious or parasitic disease will be accepted as causing a malignant neoplasm.

Example 48: I (a) Hepatocellular carcinoma
(b) Hepatitis B virus

Code to hepatocellular carcinoma (C22.0).

Example 49: I (a) Burkitt's tumour
(b) Epstein-Barr virus

Code to Burkitt's tumour (C83.7).

Example 50: I (a) Cholangiocarcinoma of liver
(b) Clonorchiasis

Code to malignant neoplasm of intrahepatic bile duct (C22.1).

J. Malignant neoplasms and circulatory disease

The following acute or fatal circulatory diseases will be accepted in Part I as due to malignant neoplasms:

I21-I22	Acute myocardial infarction
I24.-	Other acute ischaemic heart diseases
I26.-	Pulmonary embolism
I30.-	Acute pericarditis

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I33.-	Acute and subacute endocarditis
I40.-	Acute myocarditis
I44.-	Atrioventricular and left bundle-branch block
I45.-	Other conduction disorders
I46.-	Cardiac arrest
I47.-	Paroxysmal tachycardia
I48	Atrial fibrillation and flutter
I49.-	Other cardiac arrhythmias
I50.-	Heart failure
I51.8	Other ill-defined heart diseases
I60-I69	Cerebrovascular diseases, except I67.0-I67.5, I67.9, I69.-

The following circulatory diseases will not be accepted as due to malignant neoplasms:

I00-I09	Rheumatic fever and rheumatic heart disease
I10-I15	Hypertensive disease (except when reported as due to endocrine neoplasms, renal neoplasms and carcinoid tumours)
I20.-	Angina pectoris
I25.-	Chronic ischaemic heart disease
I70.-	Atherosclerosis

4.2.8 Rheumatic fever with heart involvement

If there is no statement that the rheumatic process was active at the time of death, assume activity if the heart condition (other than terminal conditions and bacterial endocarditis) that is specified as rheumatic, or stated to be due to rheumatic fever, is described as acute or subacute. In the absence of such description, the terms “carditis”, “endocarditis”, “heart disease”, “myocarditis”, and “pancarditis” can be regarded as acute if either the interval between onset and death is less than one year or, if no interval is stated, the age at death is under l5 years. “Pericarditis” can be regarded as acute at any age.

4.2.9 Congenital malformations, deformations and chromosomal abnormalities

The following conditions may be regarded as congenital when causing death at the ages stated provided there is no indication that they were acquired after birth.

- Under l year: aneurysm, aortic stenosis, atresia, atrophy of brain, cyst of brain, deformity, displacement of organ, ectopia, hypoplasia of organ, malformation, pulmonary stenosis, valvular heart disease.
- Under 4 weeks: heart disease NOS, hydrocephalus NOS.

If the interval between onset and death and the age of the decedent indicate that the condition existed from birth, any disease should be regarded as congenital even when not specified as congenital on the medical certificate.

4.2.10 Nature of injury

The codes for external causes (V01-Y89) should be used as the primary codes for single-condition coding and tabulation of mortality involving injury, poisoning and certain other consequences of external causes. It is recommended that a code from Chapter XIX (S00-T98) should be used in addition in order to identify the nature of

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the injury and permit relevant tabulations. The following notes refer to such coding. Where more than one kind of injury to a single body region in S00-S99, T08- T35, T66-T79 is mentioned and there is no clear indication as to which caused death, the General Principle and the Selection Rules should be applied in the normal way.

Example 1:

I (a) Haemorrhagic shock
(b) Peritoneal haemorrhage
(c) Rupture of liver
(d) Road traffic accident

Select rupture of liver (S36.1), since this is the starting point of the sequence terminating in the condition first entered on the certificate.

Example 2:

I (a) Fat embolism
(b) Fracture of femur
(c) Laceration of thigh
(d) Road traffic accident

Select fracture of femur (S72.9), since this is the starting point of the sequence terminating in the condition first entered on the certificate. It is "highly improbable" that laceration of the thigh would give rise to all the conditions mentioned above it.

Example 3:

I (a) Peritonitis
(b) Rupture of stomach and transverse colon
(c) Road traffic accident

Select rupture of stomach (S36.3), since this is the starting point of the first-mentioned sequence (in accordance with Rule 1).

Example 4:

I (a) Purulent meningitis
(b) Contusion of eyelid and penetrating wound of orbit

Select penetrating wound of orbit (S05.4), since contusion of eyelid selected by Rule 2 is obviously a direct consequence of the penetrating wound of the orbit (Rule 3 is applied).

When more than one body region is involved, coding should be made to the relevant category of Injuries involving multiple body regions (T00-T06). This applies both to the same type of injury and to more than one kind of injury to different body regions.

4.2.11 Poisoning by drugs, medicaments and biological substances

When combinations of medicinal agents classified differently are involved, proceed as follows: if one component of the combination is specified as the cause of death, code to that component; if no component is specified as the cause of death, code to the category provided for the combination, e.g. mixed antiepileptics (T42.5). Otherwise, if the components are classified to the same three-character category, code to the appropriate subcategory for "Other"; if not, code to T50.9.

Combinations of medicinal agents with alcohol should be coded to the medicinal agent.

4.2.12 External causes

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The codes for external causes (V01-Y89) should be used as the primary codes for single-condition coding and tabulation of the underlying cause when, and only when, the morbid condition is classifiable to Chapter XIX (Injury, poisoning and certain other consequences of external causes).

When the morbid condition is classified to Chapters I-XVIII, the morbid condition itself should be coded as the underlying cause and categories from the chapter for external causes may be used, if desired, as supplementary codes.

4.2.13 Expressions indicating doubtful diagnosis

Qualifying expressions indicating some doubt as to the accuracy of the diagnosis, such as “apparently”, “presumably”, “possibly”, etc., should be ignored, since entries without such qualification differ only in the degree of certainty of the diagnosis.

4.2.14 Human Immunodeficiency Virus (HIV)

When a blood transfusion is given as treatment for any condition (e.g. a haematological disorder) and an infected blood supply results in a HIV infection, code the HIV as the underlying cause and not the treated condition.

Example 1:

I	(a)	Kaposi's sarcoma	1 year
	(b)	HIV	3 years
	(c)	Blood transfusion	5 years
	(d)	Haemophilia	since birth

Code to HIV.

Example 2:

I	(a)	Pneumocystis carinii	6 months
	(b)	HIV	5 years
	(c)	Ruptured spleen	7 years
	(d)	Assault – fist fight	7 years

Code to HIV.

4.3 Perinatal mortality: guidelines for certification and rules for coding

4.3.1 Certification of perinatal deaths

Whenever possible, a separate certificate of cause of perinatal death should be completed, in which the causes are set out as follows:

(a) Main disease or condition in fetus or infant
(b) Other diseases or conditions in fetus or infant
(c) Main maternal disease or condition affecting fetus or infant
(d) Other maternal diseases or conditions affecting fetus or infant
(e) Other relevant circumstances

The certificate should include identifying particulars with relevant dates and times, a statement as to whether the baby was born alive or dead, and details of the autopsy.

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For a thorough analysis of perinatal mortality, the following data on both mother and child are needed, in addition to information about the causes of death, not only in the case of perinatal death, but also for all live births:

Mother
Date of birth
Number of previous pregnancies: live births/ stillbirths/ abortions
Date and outcome of last previous pregnancy: live birth/ stillbirth/ abortion Present pregnancy:

- first day of last menstrual period (if unknown, then estimated duration of pregnancy in completed weeks)
- antenatal care - two or more visits: yes/no/not known
- delivery: normal spontaneous vertex/other (specify)

Child
Birth weight in grams
Sex: boy/girl/indeterminate
Single birth/first twin/second twin/other multiple birth
If stillborn, when death occurred: before labour/during labour/not known

Other variables that might appear on the basic certificate include particulars of the birth attendant, as follows: physician/trained midwife/other trained person (specify)/other (specify).

The method by which the supplementary data are collected will vary according to the civil registration system obtaining in different countries. Where they can be collected at the registration of the stillbirth or early neonatal death, a form similar to the "Certificate of Cause of Perinatal Death" as shown below could be used. Otherwise, special arrangements would need to be made (for example, by linking birth and death records) to bring together the supplementary data and the cause of death.

Where civil registration requirements make it difficult to introduce a common death certificate for liveborn and stillborn infants, the problem could be met by separate certificates for stillbirths and early neonatal deaths, each incorporating the recommended format for the causes of death.

4.3.2 Statement of causes of death

The certificate has five sections for the entry of causes of perinatal deaths, labelled (a) to (e). In sections (a) and (b) should be entered diseases or conditions of the infant or fetus, the single most important in section (a) and the remainder, if any, in section (b). By "the single most important" is meant the pathological condition that, in the opinion of the certifier, made the greatest contribution to the death of the infant or fetus. The mode of death, e.g. heart failure, asphyxia or anoxia, should not be entered in section (a) unless it was the only fetal or infant condition known. This also holds true for prematurity.

In sections (c) and (d) should be entered all diseases or conditions of the mother that, in the certifier's opinion, had some adverse effect on the infant or fetus. Again, the most important one of these should be entered in section (c) and the others, if any, in section (d). Section (e) is for the reporting of any other circumstances that have a bearing on the death but cannot be described as a disease or condition of the infant or mother, e.g. delivery in the absence of an attendant.

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CERTIFICATE OF CAUSE OF PERINATAL DEATH

To be completed for stillbirths and liveborn infants dying within 168 hours (1 week) from birth

Identifying particulars

☐ This child was born live on at hours
and died on at hours

☐ This child was stillborn on at hours
and died before labour ☐ during labour ☐ not known ☐

Mother

Date of birth ☐☐☐☐
or, if unknown, age (years) ☐☐

Number of previous pregnancies:
Live births ☐☐
Stillbirths ☐☐
Abortions ☐☐

Outcome of last previous pregnancy:
☐ Live birth
☐ Stillbirth
☐ Abortion
Date ☐☐☐☐

1st day of last menstrual period ☐☐☐☐
or, if unknown, estimated duration of pregnancy ☐☐
(completed weeks)

Antenatal care, two or more visits:
☐ Yes
☐ No
☐ Not known

Delivery:
☐ Normal spontaneous vertex
Other (specify)
. . . . . .

Child

Birthweight: . . . . . grams

Sex:
☐ Boy ☐ Girl ☐ Indeterminate

☐ Single birth ☐ First twin
☐ Second twin ☐ Other multiple

Attendant at birth

☐ Physician ☐ Trained midwife
Other trained person (specify)
. . . . . .
Other (specify)
. . . . . .

Causes of death

a. Main disease or condition in fetus or infant

b. Other diseases or conditions in fetus or infant

c. Main maternal disease or condition affecting fetus or infant

d. Other maternal diseases or conditions affecting fetus or infant

e. Other relevant circumstances

☐ The certified cause of death has been confirmed | Identify:

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The following examples illustrate the statement of the causes of death for the cases described.

Example 1. A woman, whose previous pregnancies had ended in spontaneous abortions at 12 and 18 weeks, was admitted when 24 weeks pregnant, in premature labour. There was spontaneous delivery of a 700 g infant who died during the first day of life. The main finding at autopsy was "pulmonary immaturity".

Causes of perinatal death:

(a) Pulmonary immaturity
(b) —
(c) Premature labour, cause unknow
(d) Recurrent aborter
(e) —

Example 2. A primigravida aged 26 years with a history of regular menstrual cycles received routine antenatal care starting at the 10th week of pregnancy. At 30-32 weeks, fetal growth retardation was noted clinically, and confirmed at 34 weeks. There was no evident cause apart from symptomless bacteriuria. A caesarean section was performed and a liveborn boy weighing 1600 g was delivered. The placenta weighed 300 g and was described as infarcted. Respiratory distress syndrome developed which was responding to treatment. The baby died suddenly on the third day. Autopsy revealed extensive pulmonary hyaline membrane and massive intraventricular haemorrhage.

Causes of perinatal death:

(a) Intraventricular haemorrhage
(b) Respiratory distress syndrome Retarded fetal growth
(c) Placental insufficiency
(d) Bacteriuria in pregnancy Caesarean section

Example 3. A known diabetic, who was poorly controlled during her first pregnancy, developed megaloblastic anaemia at 32 weeks. Labour was induced at 38 weeks. There was spontaneous delivery of an infant weighing 3200 g. The baby developed hypoglycaemia and died on the second day. Autopsy showed truncus arteriosus.

Causes of perinatal death:

(a) Truncus arteriosus
(b) Hypoglycaemia
(c) Diabetes
(d) Megaloblastic anaemia
(e) —

Example 4. A 30-year-old mother of a healthy four-year-old boy had a normal pregnancy apart from hydramnios. X-ray at 36 weeks suggested anencephaly. Labour was induced. A stillborn anencephalic fetus weighing 1500 g was delivered.

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Causes of perinatal death:

(a) Anencephaly
(b) —
(c) Hydramnios
(d) —
(e) —

4.3.3 Tabulation of perinatal mortality by cause

For statistics of perinatal mortality derived from the form of certificate shown in the accompanying documentation, full-scale multiple-cause analysis of all conditions reported will yield the maximum benefit. Where this is impracticable, analysis of the main disease or condition in the fetus or infant (part (a)) and of the main maternal condition affecting the fetus or infant (part (c)) with cross-tabulation of groups of these conditions should be regarded as the minimum. Where it is necessary to select only one condition (for example, when it is necessary to incorporate early neonatal deaths in singlecause tables of deaths at all ages), the main disease or condition in the fetus or infant (part (a)) should be selected.

4.3.4 Coding of causes of death

Each condition entered in sections (a), (b), (c) and (d) should be coded separately. Maternal conditions affecting the infant or fetus, entered in sections (c) and (d), should be coded to categories P00-P04 and these codes should not be used for sections (a) and (b). Conditions in the infant or fetus, entered in section (a), can be coded to any categories other than P00-P04 but will most often be coded to categories P05-P96 (Perinatal conditions) or Q00-Q99 (Congenital anomalies). Only one code should be entered for sections (a) and (c), but for sections (b) and (d) as many codes should be entered as there are conditions reported.

Section (e) is for review of individual perinatal deaths and will not normally need to be coded. If, however, a statistical analysis of the circumstances entered in section (e) is desired, some suitable categories may exist in Chapters XX and XXI; where this is not the case, users should devise their own coding system for this information.

4.3.5 Coding rules

The selection rules for general mortality do not apply to the perinatal death certificate. It may happen, however, that perinatal death certificates are received on which the causes of death have not been entered in accordance with the guidelines given above. Whenever possible, these certificates should be corrected by the certifier, but if this is not possible, the following rules should be applied.

Rule P1. Mode of death or prematurity entered in section (a).

If heart or cardiac failure, asphyxia or anoxia (any condition in P20.-, P21.-) or prematurity (any condition in P07.-) is entered in section (a) and other conditions of the infant or fetus are entered in either section (a) or section (b), code the first-mentioned of these other conditions as if it had been entered alone in section (a) and code the condition actually entered in section (a) as if it had been entered in section (b).

Example 1:	Liveborn; death at 4 days	Coding
	(a) Prematurity	Q05.9

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(b)	Spina bifida	P07.3
(c)	Placental insufficiency	P02.2
(d)	—	

Prematurity is coded at (b) and spina bifida at (a).

Example 2: Liveborn; death at 50 minutes — Coding

(a)	Severe birth asphyxia Hydrocephalus	Q03.9
(b)	—	P21.0
(c)	Obstructed labour	P03.1
(d)	Severe pre-eclampsia	P00.0

Severe birth asphyxia is coded at (b) and hydrocephalus at (a).

Rule P2. Two or more conditions entered in sections (a) or (c).

If two or more conditions are entered in section (a) or section (c), code the first-mentioned of these as if it had been entered alone in section (a) or (c) and code the others as if they had been entered in sections (b) or (d).

Example 3: Stillborn; death before onset of labour — Coding

(a)	Severe fetal malnutrition Light for dates Antepartum anoxia	P05.0
(b)	—	P20.9
(c)	Severe pre-eclampsia Placenta praevia	P00.0
(d)	—	P02.0

Light for dates with fetal malnutrition is coded at (a) and antepartum anoxia at (b); severe pre-eclampsia is coded at (c) and placenta praevia at (d).

Example 4: Liveborn; death at 2 days — Coding

(a)	Traumatic subdural haemorrhage Massive inhalation of meconium Intrauterine anoxia	P10.0
(b)	Hypoglycaemia Prolonged pregnancy	P24.0 P20.9 P70.4 P08.2
(c)	Forceps delivery P03.2	
(d)	Severe pre-eclampsia P00.0	

Traumatic subdural haemorrhage is coded at (a) and the other conditions entered in (a) are coded at (b).

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Rule P3. No entry in sections (a) or (c).

If there is no entry in section (a) but there are conditions of the infant or fetus entered in section (b), code the first-mentioned of these as if it had been entered in section (a); if there are no entries in either section (a) or section (b), either code P95 (Fetal death of unspecified cause) for stillbirths or code P96.9 (Condition originating in the perinatal period, unspecified) for early neonatal deaths should be used for section (a).

Similarly, if there is no entry in section (c) but there are maternal conditions entered in section (d), code the first-mentioned of these as if it had been entered in section (c); if there are no entries in either section (c) or section (d) use some artificial code, e.g. xxx.x for section (c) to indicate that no maternal condition was reported.

Example 5:	Liveborn; death at 15 minutes	Coding
	(a) —	P10.4
	(b) Tentorial tear	P22.0
	Respiratory distress syndrome	
	(c)	xxx.x
	(d) —	

Tentorial tear is coded at (a); xxx.x is coded at (c).

Example 6:	Liveborn; death at 2 days	Coding
	(a) —	P95
	(b) —	
	(c) —	P00.0
	(d) Eclampsia (longstanding essential hypertension)	

Unspecified perinatal cause is coded at (a); eclampsia is coded at (c).

Rule P4. Conditions entered in wrong section.

If a maternal condition (i.e. conditions in P00-P04) is entered in section (a) or section (b), or if a condition of the infant or fetus is entered in section (c) or section (d), code the conditions as if they had been entered in the respective correct section.

If a condition classifiable as a condition of the infant or fetus or as a maternal condition is mistakenly entered in section (e), code it as an additional fetal or maternal condition in section (b) or (d) respectively.

Example 7:	Stillborn; death after onset of labour	Coding
	(a) Severe intrauterine hypoxia	P20.9
	(b) Persistent occipitoposterior	
	(c) —	P03.1
	(d) —	P03.2
	(e) Difficult forceps delivery	

Persistent occipitoposterior is coded at (c); difficult forceps delivery is coded at (d).

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4.4 Morbidity

At the time of the sixth revision of the ICD, adopted in 1948, a number of requests were received from public health administrators, health care managers, social security authorities and researchers in various health disciplines for a classification suitable for morbidity applications. The ICD was, therefore, made suitable for grouping morbidity data, in addition to its traditional uses, and the morbidity aspect has since been progressively expanded through successive revisions. Morbidity data are increasingly being used in the formulation of health policies and programmes, and in their management, monitoring and evaluation, in epidemiology, in identification of risk populations, and in clinical research (including studies of disease occurrence in different socioeconomic groups).

The condition to be used for single-condition morbidity analysis is the main condition treated or investigated during the relevant episode of health care. The main condition is defined as the condition, diagnosed at the end of the episode of health care, primarily responsible for the patient's need for treatment or investigation. If there is more than one such condition, the one held most responsible for the greatest use of resources should be selected. If no diagnosis was made, the main symptom, abnormal finding or problem should be selected as the main condition.

In addition to the main condition, the record should, whenever possible, also list separately other conditions or problems dealt with during the episode of health care. Other conditions are defined as those conditions that coexist or develop during the episode of health care and affect the management of the patient. Conditions related to an earlier episode that have no bearing on the current episode should not be recorded.

By limiting analysis to a single condition for each episode, some available information may be lost. It is therefore recommended, where practicable, to carry out multiple-condition coding and analysis to supplement the routine data. This should be done according to local rules, since no international rules have been established. However, experience in other areas could be useful in developing local schemes.

4.4.1 Guidelines for recording diagnostic information for single-condition analysis of morbidity data

General

The health care practitioner responsible for the patient's treatment should select the main condition to be recorded, as well as any other conditions, for each episode of health care. This information should be organized systematically by using standard recording methods. A properly completed record is essential for good patient management and is a valuable source of epidemiological and other statistical data on morbidity and other health care problems.

Specificity and detail

Each diagnostic statement should be as informative as possible in order to classify the condition to the most specific ICD category. Examples of such diagnostic statements include:

- transitional cell carcinoma of trigone of bladder
- acute appendicitis with perforation
- diabetic cataract, insulin-dependent
- meningococcal pericarditis
- antenatal care for pregnancy-induced hypertension • diplopia due to allergic reaction to antihistamine taken as prescribed

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- osteoarthritis of hip due to an old hip fracture • fracture of neck of femur following a fall at home
- third-degree burn of palm of hand.

Uncertain diagnoses or symptoms

If no definite diagnosis has been established by the end of an episode of health care, then the information that permits the greatest degree of specificity and knowledge about the condition that necessitated care or investigation should be recorded. This should be done by stating a symptom, abnormal finding or problem, rather than qualifying a diagnosis as "possible", "questionable" or "suspected", when it has been considered but not established.

Contact with health services for reasons other than illness

Episodes of health care or contact with health services are not restricted to the treatment or investigation of current illness or injury. Episodes may also occur when someone who may not currently be sick requires or receives limited care or services; the details of the relevant circumstances should be recorded as the "main condition". Examples include:

- monitoring of previously treated conditions
- immunization
- contraceptive management, antenatal and postpartum care
- surveillance of persons at risk because of personal or family history
- examinations of healthy persons, e.g. for insurance or occupational reasons
- seeking of health-related advice
- requests for advice by persons with social problems
- consultation on behalf of a third party.

Chapter XXI (Factors influencing health status and contact with health services) provides a broad range of categories (Z00-Z99) for classifying these circumstances; reference to this chapter will give an indication of the detail required to permit classification to the most relevant category.

Multiple conditions

Where an episode of health care concerns a number of related conditions (e.g. multiple injuries, multiple sequelae of a previous illness or injury, or multiple conditions occurring in human immunodeficiency virus [HIV] disease), the one that is clearly more severe and demanding of resources than the others should be recorded as the "main condition" and the others as "other conditions". Where no one condition predominates, a term such as "multiple fractures", "multiple head injuries", or "HIV disease resulting in multiple infections" may be recorded as the "main condition", followed by a list of the conditions. If there are a number of such conditions, with none predominating, then a term such as "multiple injuries" or "multiple crushing injuries" should be recorded alone.

Conditions due to external causes

When a condition such as an injury, poisoning or other effect of external causes is recorded, it is important to describe fully both the nature of the condition and the circumstances that gave rise to it. For example: "fracture of neck of femur caused by fall due to slipping on greasy pavement"; "cerebral contusion caused when patient lost control of car, which hit a tree"; "accidental poisoning - patient drank disinfectant in mistake for soft drink"; or "severe hypothermia - patient fell in her garden in cold weather".

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Treatment of sequelae

Where an episode of care is for the treatment or investigation of a residual condition (sequela) of a disease that is no longer present, the sequela should be fully described and its origin stated, together with a clear indication that the original disease is no longer present. For example: “deflected nasal septum - fracture of nose in childhood”, “contracture of Achilles tendon - late effect of injury to tendon”, or “infertility due to tubal occlusion from old tuberculosis”.

Where multiple sequelae are present and treatment or investigation is not directed predominantly at one of them, a statement such as “sequelae of cerebrovascular accident” or “sequelae of multiple fractures” is acceptable.

4.4.2 Guidelines for coding “main condition” and “other conditions”

General

The “main condition” and “other conditions” relevant to an episode of health care should have been recorded by the responsible health care practitioner, and coding is therefore usually straightforward, since the main condition stated should be accepted for coding and processing unless it is obvious that the guidelines given above have not been followed. Whenever possible, a record with an obviously inconsistent or incorrectly recorded main condition should be returned for clarification. Failing clarification, Rules MB1 to MB5 (section 4.4.3) will help the coder to deal with some of the commoner causes of incorrect recording. The guidelines given below are for use when the coder may be unclear as to which code should be used.

It has been recommended that “other conditions” in relation to an episode of care should be recorded in addition to the main condition, even for singlecause analysis, since this information may assist in choosing the correct ICD code for the main condition.

Optional additional codes

In the guidelines below, a preferred code for the “main condition” is sometimes indicated, together with an optional additional code to give more information. The preferred code indicates the “main condition” for singlecause analysis and an additional code may be included for multiple-cause analysis.

Coding of conditions to which the dagger and asterisk system applies

If applicable, both dagger and asterisk codes should be used for the main condition, since they denote two different pathways for a single condition.

Example 1: Main condition: Measles pneumonia
Other conditions: —

Code to measles complicated by pneumonia (B05.2†) and pneumonia in viral diseases classified elsewhere (J17.1*).

Example 2: Main condition: Tuberculous pericarditis
Other conditions: —

Code to tuberculosis of other specified organs (A18.8†) and pericarditis in bacterial diseases classified elsewhere (I32.0*).

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Example 3: Main condition: Lyme disease arthritis
Other conditions: —

Code to Lyme disease (A69.2†) and arthritis in Lyme disease (M01.2*).

Coding of suspected conditions, symptoms and abnormal findings and non-illness situations

If the period of health care was for an inpatient, the coder should be cautious about classifying the main condition to Chapters XVIII and XXI. If a more specific diagnosis has not been made by the end of the inpatient stay, or if there was truly no codable current illness or injury, then codes from the above chapters are permissible (see also Rules MB3 and MB5, section 4.4.3). The categories can be used in the normal way for other episodes of contact with health services.

If, after an episode of health care, the main condition is still recorded as "suspected", "questionable", etc., and there is no further information or clarification, the suspected diagnosis must be coded as if established.

Category Z03.- (Medical observation and evaluation for suspected diseases and conditions) applies to suspected diagnoses that can be ruled out after investigation.

Example 4: Main condition: Suspected acute cholecystitis
Other conditions: —

Code to acute cholecystitis (K81.0) as "main condition".

Example 5: Main condition: Admitted for investigation of suspected malignant neoplasm of cervix - ruled out
Code to observation for suspected malignant neoplasm (Z03.1) as "main condition".

Example 6: Main condition: Ruled out myocardial infarction
Other conditions: —

Code to observation for suspected myocardial infarction (Z03.4) as "main condition".

Example 7: Main condition: Severe epistaxis
Other conditions: —

Patient in hospital one day. No procedures or investigations reported

Code to epistaxis (R04.0). This is acceptable since the patient was obviously admitted to deal with the immediate emergency only.

Coding of multiple conditions

Where multiple conditions are recorded in a category entitled "Multiple ...", and no single condition predominates, the code for the "Multiple ..." category should be used as the preferred code, and optional additional codes may be added for individual conditions listed.

Such coding applies mainly to conditions associated with HIV disease, to injuries and sequelae.

Coding of combination categories

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The ICD provides certain categories where two conditions or a condition and an associated secondary process can be represented by a single code. Such combination categories should be used as the main condition where appropriate information is recorded. The Alphabetical Index indicates where such combinations are provided for, under the indent "with", which appears immediately after the lead term. Two or more conditions recorded under "main condition" may be linked if one of them may be regarded as an adjectival modifier of the other.

Example 8:	Main condition:	Renal failure
	Other conditions:	Hypertensive renal disease

Code to hypertensive renal disease with renal failure (I12.0) as the "main condition".

Example 9:	Main condition:	Glaucoma secondary to eye inflammation
	Other conditions:	—

Code to glaucoma secondary to eye inflammation (H40.4) as the "main condition".

Example 10:	Main condition:	Intestinal obstruction
	Other conditions:	Left inguinal hernia

Code to unilateral or unspecified inguinal hernia, with obstruction, without gangrene (K40.3).

Example 11:	Main condition:	Cataract. Insulin-dependent diabetes
	Other conditions:	Hypertension
	Specialty:	Ophthalmology

Code to insulin-dependent diabetes with ophthalmic complications (E10.3†) and diabetic cataract (H28.0*) as the "main condition".

Example 12:	Main condition:	Non-insulin-dependent diabetes mellitus
	Other conditions:	Hypertension
		Rheumatoid arthritis
		Cataract
	Specialty:	General medicine

Code to non-insulin-dependent diabetes without complications (E11.9) as "main condition". Note that in this example the linkage of cataract with diabetes must not be made since they are not both recorded under "main condition".

Coding of external causes of morbidity

For injuries and other conditions due to external causes, both the nature of the condition and the circumstances of the external cause should be coded. The preferred "main condition" code should be that describing the nature of the condition. This will usually, but not always, be classifiable to Chapter XIX. The code from Chapter XX indicating the external cause would be used as an optional additional code.

Example 13:	Main condition:	Fracture of neck of femur caused by fall due to tripping on uneven pavement
	Other conditions:	Contusions to elbow and upper arm

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Code to fracture of neck of femur (S72.0) as “main condition”. The external cause code for fall on same level from slipping, tripping or stumbling on street or highway (W01.4) may be used as an optional additional code.

Example 14: Main condition: Severe hypothermia - patient fell in her garden in cold weather
Other conditions: Senility

Code to hypothermia (T68) as “main condition”. The external cause code for exposure to excessive natural cold at home (X31.0) may be used as an optional additional code.

Example 15: Main condition: Diplopia due to allergic reaction to antihistamine taken as prescribed
Other conditions: —

Code to diplopia (H53.2) as the “main condition”. The external cause code for antiallergic and antiemetic drugs causing adverse effects in therapeutic use (Y43.0) may be used as an optional additional code.

Example 16: Main condition: Haemoglobinuria caused by training for marathon run (training on outdoor track at stadium)
Other conditions: —

Code to haemoglobinuria due to haemolysis from other external causes (D59.6) as “main condition”. The external cause code for overexertion and strenuous, repetitive movements at sports and athletics area (X50.3) may be used as an optional additional code.

Coding of sequelae of certain conditions

The ICD provides a number of categories entitled “Sequelae of ...” (B90- B94, E64.-, E68, G09, I69.-, O97, T90-T98, Y85-Y89) which may be used to indicate conditions no longer present as the cause of a current problem undergoing treatment or investigation. The preferred code for the “main condition” is, however, the code for the nature of the sequela itself, to which the code for “Sequelae of ...” may be added as an optional additional code. Where a number of different very specific sequelae are present and no one of them predominates in severity and use of resources for treatment, it is permissible for the description “Sequelae of ...” to be recorded as the “main condition” and this may then be coded to the appropriate category. Note that it is sufficient that the causal condition is described as “old”, “no longer present”, etc. or the resulting condition is described as “late effect of ...”, or “sequela of ...” for this to apply. There is no minimum time interval.

Example 17: Main condition: Dysphasia from old cerebral infarction
Other conditions: —

Code to dysphasia (R47.0) as the “main condition”. The code for sequelae of cerebral infarction (I69.3) may be used as an optional additional code.

Example 18: Main condition: Osteoarthritis of hip joint due to old hip fracture from motor vehicle accident 10 years ago
Other conditions: —

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Code to other post-traumatic coxarthrosis (M16.5) as the "main condition". The codes for sequelae of fracture of femur (T93.1) and sequelae of motor vehicle accident (Y85.0) may be used as optional additional codes.

Example 19: Main condition: Late effects of poliomyelitis
Other conditions: —

Code to sequelae of poliomyelitis (B91) as the "main condition" since no other information is available.

Coding of acute and chronic conditions

Where the main condition is recorded as being both acute (or subacute) and chronic, and the ICD provides separate categories or subcategories for each, but not for the combination, the category for the acute condition should be used as the preferred main condition.

Example 20: Main condition: Acute and chronic cholecystitis

Other conditions: —

Code to acute cholecystitis (K81.0) as the "main condition". The code for chronic cholecystitis (K81.1) may be used as an optional additional code.

Example 21: Main condition: Acute exacerbation of chronic obstructive bronchitis

Other conditions: —

Code to chronic obstructive pulmonary disease with acute exacerbation (J44.1) as the "main condition" since the ICD provides an appropriate code for the combination.

Coding of postprocedural conditions and complications

Categories are provided in Chapter XIX (T80-T88) for certain complications related to surgical and other procedures, e.g. surgical wound infections, mechanical complications of implanted devices, shock, etc. Most bodysystem chapters also contain categories for conditions that occur either as a consequence of specific procedures and techniques or as a result of the removal of an organ, e.g. postmastectomy lymphoedema syndrome, postirradiation hypothyroidism. Some conditions (e.g. pneumonia, pulmonary embolism) that may arise in the postprocedural period are not considered unique entities and are, therefore, coded in the usual way, but an optional additional code from Y83-Y84 may be added to identify the relationship to a procedure.

When postprocedural conditions and complications are recorded as the main condition, reference to modifiers or qualifiers in the Alphabetical Index is essential for choosing the correct code.

Example 22: Main condition: Hypothyroidism since thyroidectomy 1 year ago
Other conditions: —
Specialty: General medicine

Code to postsurgical hypothyroidism (E89.0) as the "main condition".

Example 23: Main condition: Excessive haemorrhage after tooth extraction

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	Other conditions:	Pain
	Specialty:	Dentistry

Code to haemorrhage resulting from a procedure (T81.0) as the "main condition".

Example 24:	Main condition:	Postoperative psychosis after plastic surgery
	Other conditions:	—
	Specialty:	Psychiatry

Code to psychosis (F09) as the "main condition" and supplement by Y83.8 (other specified surgical procedures [as the cause of abnormal reaction of the patient]) to indicate the postprocedural relationship.

4.4.3 Rules for reselection when the main condition is incorrectly recorded

The responsible health care practitioner indicates the "main condition" to be coded, and this should normally be accepted for coding subject to the guidelines above and in the chapter-specific notes in section 4.4.4. However, certain circumstances or the availability of other information may indicate that the health care practitioner has not followed the correct procedure. If it is not possible to obtain clarification from the health care practitioner, one of the following rules may be applied and the "main condition" reselected.

Rules for reselection of main condition

Rule MB1. Minor condition recorded as "main condition", more significant condition recorded as "other condition"

Where a minor or longstanding condition, or an incidental problem, is recorded as the "main condition", and a more significant condition, relevant to the treatment given and/or the specialty that cared for the patient, is recorded as an "other condition", reselect the latter as the "main condition".

Rule MB2. Several conditions recorded as "main condition".

If several conditions that cannot be coded together are recorded as the "main condition", and other details on the record point to one of them as the "main condition" for which the patient received care, select that condition. Otherwise select the condition first mentioned.

Rule MB3. Condition recorded as "main condition" is presenting symptom of diagnosed, treated condition

If a symptom or sign (usually classifiable to Chapter XVIII), or a problem classifiable to Chapter XXI, is recorded as the "main condition" and this is obviously the presenting sign, symptom or problem of a diagnosed condition recorded elsewhere and care was given for the latter, reselect the diagnosed condition as the "main condition".

Rule MB4. Specificity

Where the diagnosis recorded as the "main condition" describes a condition in general terms, and a term that provides more precise information about the site or nature of the condition is recorded elsewhere, reselect the latter as the "main condition".

Rule MB5. Alternative main diagnoses

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Where a symptom or sign is recorded as the "main condition" with an indication that it may be due to either one condition or another, select the symptom as the "main condition". Where two or more conditions are recorded as diagnostic options for the "main condition", select the first condition recorded.

Examples of application of the rules for reselection of main condition

Rule MB1. Minor condition recorded as "main condition", more significant condition recorded as "other condition"

Where a minor or longstanding condition, or an incidental problem, is recorded as the "main condition", and a more significant condition, relevant to the treatment given and/or the specialty that cared for the patient, is recorded as an "other condition", reselect the latter as the "main condition".

Example 1:	Main condition:	Acute sinusitis
	Other conditions:	Carcinoma of endocervix Hypertension Patient in hospital for three weeks
	Procedure:	Total hysterectomy
	Specialty:	Gynaecology

Reselect carcinoma of endocervix as the "main condition" and code to C53.0.

Example 2:	Main condition:	Rheumatoid arthritis
	Other conditions:	Diabetes mellitus Strangulated femoral hernia Generalized arteriosclerosis Patient in hospital for two weeks
	Procedure:	Herniorrhaphy Specialty: Surgery

Reselect strangulated femoral hernia as the "main condition" and code to K41.3.

Example 3:	Main condition:	Epilepsy
	Other conditions:	Otomycosis
	Specialty:	Ear, nose and throat

Reselect otomycosis as the "main condition" and code to B36.9† and H62.2*.

Example 4:	Main condition:	Congestive heart failure
	Other conditions:	Fracture neck of femur due to fall from bed during hospitalization Patient in hospital for four weeks
	Procedure:	Internal fixation of fracture Specialty: Internal medicine for 1 week then transfer to orthopaedic surgery for treatment of fracture

Reselect fracture of neck of femur as the "main condition" and code to S72.0.

Example 5:	Main condition:	Dental caries
	Other conditions:	Rheumatic mitral stenosis
	Procedure:	Dental extractions

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Specialty: Dentistry

Select dental caries as the "main condition" and code to K02.9. Rule MB1 does not apply. Although dental caries can be regarded as a minor condition and rheumatic mitral stenosis as a more significant condition, the latter was not the condition treated during the episode of care.

Rule MB2. Several conditions recorded as "main condition"

If several conditions that cannot be coded together are recorded as the "main condition", and other details on the record point to one of them as being the "main condition" for which the patient received care, select that condition. Otherwise select the condition first mentioned.

Note: See also 4.4.2, coding of multiple conditions and coding of combination categories.

Example 6:

Main condition: Cataract
Staphylococcal meningitis
Ischaemic heart disease
Other conditions: —
Patient in hospital for five weeks
Specialty: Neurology

Select staphylococcal meningitis as the "main condition" and code to G00.3.

Example 7:

Main condition: Chronic obstructive bronchitis
Hypertrophy of prostate
Psoriasis vulgaris

Outpatient in the care of a dermatologist

Select psoriasis vulgaris as the "main condition" and code to L40.0.

Example 8:

Main condition: Mitral stenosis
Acute bronchitis
Rheumatoid arthritis
Other conditions: —
Specialty: General medicine
No information about therapy

Select mitral stenosis, the first-mentioned condition, as the "main condition" and code to I05.0.

Example 9:

Main condition: Chronic gastritis
Secondary malignancy in axillary lymph nodes
Carcinoma of breast
Other conditions: —
Procedure: Mastectomy

Select malignant neoplasm of breast as the "main condition" and code to C50.9.

Example 10:

Main condition: Premature rupture of membranes

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	Breech presentation
	Anaemia
Other conditions:	—
Procedure:	Spontaneous delivery

Select premature rupture of membranes, the first-mentioned condition, as the "main condition" and code to O42.9.

Rule MB3. Condition recorded as "main condition" is presenting symptom of diagnosed, treated condition

If a symptom or sign (usually classifiable to Chapter XVIII), or a problem classifiable to Chapter XXI, is recorded as the "main condition" and this is obviously the presenting sign, symptom or problem of a diagnosed condition recorded elsewhere and care was given for the latter, reselect the diagnosed condition as the "main condition".

Example 11:

Main condition:	Haematuria
Other conditions:	Varicose veins of legs
	Papillomata of posterior wall of bladder
Treatment:	Diathermy excision of papillomata
Specialty:	Urology

Reselect papillomata of posterior wall of bladder as the "main condition" and code to D41.4.

Example 12:

Main condition:	Coma
Other conditions:	Ischaemic heart disease
	Otosclerosis
	Diabetes mellitus, insulin-dependent
Specialty:	Endocrinology
Care:	Establishment of correct dose of insulin

Reselect diabetes mellitus, insulin-dependent as the "main condition" and code to E10.0. The information provided indicates that the coma was due to diabetes mellitus and coma is taken into account as it modifies the coding.

Example 13:

Main condition:	Abdominal pain
Other conditions:	Acute appendicitis
Procedure:	Appendectomy

Reselect acute appendicitis as the "main condition" and code to K35.9.

Example 14:

Main condition:	Febrile convulsions
Other conditions:	Anaemia

No information about therapy

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Accept febrile convulsions as the “main condition” and code to R56.0. Rule MB3 does not apply since the “main condition” as reported is not a presenting symptom of the other reported condition.

Rule MB4. Specificity

Where the diagnosis recorded as the “main condition” describes a condition in general terms, and a term that provides more precise information about the site or nature of the condition is recorded elsewhere, reselect the latter as the “main condition”.

Example 15:

Main condition:	Cerebrovascular accident
Other conditions:	Diabetes mellitus
	Hypertension
	Cerebral haemorrhage

Reselect cerebral haemorrhage as the “main condition” and code to I61.9.

Example 16:

Main condition:	Congenital heart disease
Other conditions:	Ventricular septal defect

Reselect ventricular septal defect as the “main condition” and code to Q21.0.

Example 17:

Main condition:	Enteritis
Other conditions:	Crohn's disease of ileum

Reselect Crohn's disease of ileum as the “main condition” and code to K50.0.

Example 18:

Main condition:	Dystocia
Other conditions:	Hydrocephalic fetus
	Fetal distress
Procedure:	Caesarean section

Reselect obstructed labour due to other abnormalities of fetus as the “main condition” and code to O66.3.

Rule MB5. Alternative main diagnoses

Where a symptom or sign is recorded as the “main condition” with an indication that it may be due to either one condition or another, select the symptom as the “main condition”. Where two or more conditions are recorded as diagnostic options for the “main condition”, select the first condition recorded.

Example 19:

Main condition:	Headache due to either stress and tension or acute sinusitis
Other conditions:	—

Select headache as the “main condition” and code to R51.

Example 20:

Main condition:	Acute cholecystitis or acute pancreatitis
Other conditions:	—

Select acute cholecystitis as the “main condition” and code to K81.0.

Example 21:

Main condition:	Gastroenteritis due to infection or food poisoning

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Other conditions: —

Select infectious gastroenteritis as the "main condition" and code to A09.

4.4.4 Chapter-specific notes

Guidance is given below for specific chapters where problems may be encountered in selecting preferred "main condition" codes. The preceding general guidelines and rules apply to all chapters unless a specific chapter note states otherwise.

Chapter I: Certain infectious and parasitic diseases

B20-B24 Human immunodeficiency virus [HIV] disease

A patient with a compromised immune system due to HIV disease may sometimes require treatment during the same episode of care for more than one disease, for example mycobacterial and cytomegalovirus infections. Categories and subcategories are provided in this block for HIV disease with various other resultant diseases. Code the appropriate subcategory for the "main condition" as selected by the health care practitioner.

Where the "main condition" has been recorded as HIV disease with multiple accompanying diseases, the appropriate .7 subcategory from B20-B22 should be coded. Conditions classifiable to two or more subcategories of the same category should be coded to the .7 subcategory of the relevant category (e.g. B20 or B21). Subcategory B22.7 should be used when conditions classifiable to two or more categories from B20-B22 are present. Additional codes from within the block B20-B24 may be used, if desired, to specify the individual conditions listed.

In those rare instances when the associated condition clearly predates the HIV infection, the combination should not be coded and the selection rules should be followed.

Example 1: Main condition: HIV disease and Kaposi's sarcoma
Other conditions: —

Code to HIV disease resulting in Kaposi's sarcoma (B21.0).

Example 2: Main condition: Toxoplasmosis and cryptococcosis in HIV patient
Other conditions: —

Code to HIV disease resulting in multiple infections (B20.7). B20.8 (HIV disease resulting in other infectious and parasitic diseases) and B20.5 (HIV disease resulting in other mycoses) may be used as additional codes, if desired.

Example 3: Main condition: HIV disease with *Pneumocystis carinii* pneumonia, Burkitt's lymphoma and oral candidiasis
Other conditions: —

Code to HIV disease resulting in multiple diseases (B22.7). Additional codes B20.6 (HIV disease resulting in Pneumocystis carinii pneumonia), B21.1 (HIV disease resulting in Burkitt's lymphoma) and B20.4 (HIV disease resulting in candidiasis) may be used, if desired.

The subcategories at B20-B23 are the only optional four-character codes for countries using the four-character version of ICD-10. Where it is not desired to use these optional fourth-character subcategories, codes from

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elsewhere in the classification should be used as additional codes to identify the specific resultant conditions. In Example 1 above, the "main condition" would be coded to B21 (HIV disease resulting in malignant neoplasm). Code C46.9 (Kaposi's sarcoma) would be used as an additional code. In Example 2, the "main condition" would be coded to B20 (HIV disease resulting in infectious and parasitic diseases). Codes B58.9 (Toxoplasmosis, unspecified) and B45.9 (Cryptococcosis, unspecified) would be used as additional codes.

Whether to use the four-character subcategories of B20-B23 or multiplecause coding to identify the specific conditions is a policy decision which should be made at the time ICD-10 is implemented.

B90-B94 Sequelae of infectious and parasitic diseases

These codes are not to be used as the preferred codes for "main condition" if the nature of the residual condition is recorded. When coding to the residual condition, B90-B94 may be used as optional additional codes (see 4.4.2, Coding of sequelae of certain conditions).

B95-B97 Bacterial, viral and other infectious agents

These codes are not to be used as "main condition" codes. The categories are provided for optional use as additional codes to identify the infectious agent or organism in diseases classified outside Chapter I. Infections of unspecified site due to these agents are classified elsewhere in Chapter I.

Example 4: Main condition: Acute cystitis due to *E. coli*
Other conditions: —

Code to acute cystitis (N30.0) as the "main condition", B96.2 (*E. coli* as the cause of diseases classified to other chapters) may be used as an optional additional code.

Example 5: Main condition: Bacterial infection
Other conditions: —

Code to bacterial infection, unspecified (A49.9), as the "main condition", not to a code from B95-B97.

Chapter II: Neoplasms

When coding neoplasms, refer to the notes introducing Chapter II in Volume 1 and to the introduction of the Alphabetical Index (Volume 3) regarding code assignment and the use of morphological descriptions.

A neoplasm, whether primary or metastatic, that is the focus of care during a relevant episode of health care, should be recorded and coded as the "main condition". When the "main condition" as recorded by the health care practitioner is a primary neoplasm that is no longer present (having been removed during a previous episode of care), code as the "main condition" the neoplasm of the secondary site, the current complication, or the appropriate circumstance codable to Chapter XXI (see 4.4.1, Contact with health services for reasons other than illness) that was the focus of the treatment or investigation during the current episode of care. An appropriate code from Chapter XXI for personal history of neoplasm may be used as an optional additional code.

Example 6: Main condition: Carcinoma of prostate
Other conditions: Chronic bronchitis
Procedure: Prostatectomy

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Code to malignant neoplasm of prostate (C61) as the "main condition".

Example 7: Main condition: Carcinoma of breast - resected two years ago
Other conditions: Secondary carcinoma in lung
Procedure: Bronchoscopy with biopsy

Code to secondary malignant neoplasm of lung (C78.0) as the "main condition". Z85.3 (Personal history of malignant neoplasm of breast) may be used as an optional additional code.

Example 8: Main condition: Previously excised bladder cancer - admitted for follow-up examination by cystoscopy
Other conditions: —
Procedure: Cystoscopy

Code to follow-up examination after surgery for malignant neoplasm (Z08.0) as the "main condition". Z85.5 (Personal history of malignant neoplasm of urinary tract) may be used as an optional additional code.

C80 Malignant neoplasm without specification of site

C97 Malignant neoplasms of independent (primary) multiple sites

C80 should be used for "main condition" coding only when the health care practitioner has clearly recorded the neoplasm in such a manner. C97 should be used when the health care practitioner records as the "main condition" two or more independent primary malignant neoplasms, none of which predominates. Additional codes may be used to identify the individual malignant neoplasms listed.

Example 9: Main condition: Carcinomatosis
Other conditions: —

Code to malignant neoplasm without specification of site (C80).

Example 10: Main condition: Multiple myeloma and primary adenocarcinoma of prostate

Code to malignant neoplasms of independent (primary) multiple sites (C97). C90.0 (Multiple myeloma) and C61 (Malignant neoplasm of prostate) may be used as optional additional codes.

Chapter III: Diseases of the blood and blood-forming organs and certain disorders involving the immune mechanism

Certain conditions classifiable to this chapter may result from drugs or other external causes. Codes from Chapter XX may be used as optional additional codes.

Example 11: Main condition: Trimethoprim-induced folate deficiency anaemia
Other conditions: —

Code drug-induced folate deficiency anaemia (D52.1) as the "main condition". Y41.2 (Antimalarials and drugs acting on other blood protozoa causing adverse effects in therapeutic use) may be used as an optional additional code.

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Chapter IV: Endocrine, nutritional and metabolic diseases

Certain conditions classifiable to this chapter may result from drugs or other external causes. Codes from Chapter XX may be used as optional additional codes.

E10-E14 Diabetes mellitus

In coding the "main condition", the selection of an appropriate subcategory from the list that applies to all of these categories should be based on the "main condition" as recorded by the health care practitioner. The subcategory .7 should be used as the "main condition" code only when multiple complications of diabetes have been recorded as the "main condition" without preference for any one complication. Codes for any individual complications listed may be added as optional additional codes.

Example 12: Main condition: Renal failure due to diabetic glomerulonephrosis
Code to unspecified diabetes mellitus with renal complications (E14.2† and N08.3*).

Example 13: Main condition: Insulin-dependent diabetic with nephropathy, gangrene and cataracts
Other conditions: —

Code to insulin-dependent diabetes mellitus with multiple complications (E10.7). Codes E10.2† and N08.3* (Insulindependent diabetes with nephropathy), E10.5 (Insulindependent diabetes with peripheral circulatory complications) and E10.3† and H28.0* (Insulin-dependent diabetes with cataract) may be added as optional additional codes to identify the individual complications.

E34.0 Carcinoid syndrome

This code is not to be used as the preferred code for the "main condition" if a carcinoid tumour is recorded, unless the episode of care was directed predominantly at the endocrine syndrome itself. When coding to the tumour, E34.0 may be used as an optional additional code to identify the functional activity.

E64.- Sequelae of malnutrition and other nutritional deficiencies

E68 Sequelae of hyperalimentation

These codes are not to be used as the preferred code for the "main condition" if the nature of the residual condition is recorded. When coding to the residual condition, E64.- or E68 may be used as an optional additional code.

Chapter V: Mental and behavioural disorders

The definitions of the categories and subcategories in this chapter are provided to assist the health care practitioner in establishing diagnostic labels; they should not be used by coders. The "main condition" code should be assigned on the basis of the diagnosis recorded by the practitioner, even if there appears to be a conflict between the condition as recorded and the definition. In some categories there is provision for optional additional codes.

Chapter VI: Diseases of the nervous system

Certain conditions classifiable to this chapter may result from the effects of drugs or other external causes. Codes from Chapter XX may be used as optional additional codes.

G09 Sequelae of inflammatory diseases of central nervous system

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This code is not to be used as the preferred code for the "main condition" if the nature of the residual condition is recorded. When coding to the residual condition, G09 may be used as an optional additional code. Note that sequelae of categories G01*, G02*, G05* and G07* should not be assigned to G09, but rather to the categories established for sequelae of the underlying condition, e.g. B90-B94. If there is no sequelae category for the underlying condition, code to the underlying condition itself.

Example 14: Main condition: Deafness due to tuberculous meningitis
Specialty: Speech and hearing clinic

Code hearing loss, unspecified (H91.9) as the "main condition". B90.0 (Sequelae of central nervous system tuberculosis) may be used as an optional additional code.

Example 15: Main condition: Epilepsy due to old brain abscess
Specialty: Neurology

Code epilepsy, unspecified (G40.9) as the "main condition". G09 (Sequelae of inflammatory diseases of central nervous system) may be used as an optional additional code.

Example 16: Main condition: Mild mental retardation after postimmunization encephalitis
Specialty: Psychiatry

Code mild mental retardation (F70.9) as the "main condition". G09 (Sequelae of inflammatory diseases of central nervous system) may be used as an optional additional code.

G81-G83 Paralytic syndromes

These codes are not to be used as the preferred code for the "main condition" if a current cause is recorded, unless the episode of care was mainly for the paralysis itself. When coding to the cause, G81-G83 may be used as optional additional codes.

Example 17: Main condition: Cerebrovascular accident with hemiplegia
Other conditions: —
Specialty: Neurology

Code stroke, not specified as haemorrhage or infarction (I64) as "main condition". G81.9 (Hemiplegia, unspecified) may be used as an optional additional code.

Example 18: Main condition: Cerebral infarction three years ago
Other conditions: Paralysis of left leg
Patient receiving physical therapy

Code monoplegia of lower limb (G83.1) as "main condition". I69.3 (Sequelae of cerebral infarction) may be used as an optional additional code.

Chapter VII: Diseases of the eye and adnexa

H54.- Blindness and low vision

This code is not to be used as the preferred code for the "main condition" if the cause is recorded, unless the episode of care was mainly for the blindness itself. When coding to the cause, H54.- may be used as an optional additional

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code.

Chapter VIII: Diseases of the ear and mastoid process

H90-H91 Hearing loss

These codes are not to be used as the preferred code for the "main condition" if the cause is recorded, unless the episode of care was mainly for the hearing loss itself. When coding to the cause, H90.- or H91.- may be used as an optional additional code.

Chapter IX: Diseases of the circulatory system

I15.- Secondary hypertension

This code is not to be used as the preferred code for the "main condition" if the cause is recorded, unless the episode of care was mainly for the hypertension. When coding to the cause, I15.- may be used as an optional additional code.

I69.- Sequelae of cerebrovascular disease

This code is not to be used as the preferred code for the "main condition" if the nature of the residual condition is recorded. When coding to the residual condition, I69.- may be used as an optional additional code.

Chapter XV: Pregnancy, childbirth and the puerperium

O08.- Complications following abortion and ectopic and molar pregnancy

This code is not to be used as the preferred code for the "main condition", except where a new episode of care is solely for treatment of a complication, e.g. a current complication of a previous abortion. It may be used as an optional additional code with categories O00-O02 to identify associated complications and with categories O03-O07 to give fuller details of the complication.

Note that the inclusion terms provided at the subcategories of O08 should be referred to when assigning the fourth-character subcategories of O03-O07.

Example 19: Main condition: Ruptured tubal pregnancy with shock
Specialty: Gynaecology

Code ruptured tubal pregnancy (O00.1) as the "main condition". O08.3 (Shock following abortion and ectopic and molar pregnancy) may be used as an optional additional code.

Example 20: Main condition: Incomplete abortion with perforation of uterus
Specialty: Gynaecology

Code incomplete abortion with other and unspecified complications (O06.3) as the "main condition". Code O08.6 (Damage to pelvic organs and tissues following abortion and ectopic and molar pregnancy) may be added as an optional additional code.

Example 21: Main condition: Disseminated intravascular coagulation following abortion performed two days ago at another facility
Specialty: Gynaecology

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Code delayed or excessive haemorrhage following abortion and ectopic and molar pregnancy (O08.1). No other code is required since the abortion was performed during a previous episode of care.

O80-O84 Delivery

Use of these codes to describe the "main condition" should be limited to cases where the only information recorded is a statement of delivery or the method of delivery. Codes O80-O84 may be used as optional additional codes to indicate a method or type of delivery where no separate data item or procedural classification is being used for this purpose.

Example 22:

Main condition:	Pregnancy
Other conditions:	—
Procedure:	Low forceps delivery

Code low forceps delivery (O81.0) as "main condition" since no other information is provided.

Example 23:

Main condition:	Pregnancy delivered
Other conditions:	Failed trial of labour
Procedure:	Caesarean section

Code failed trial of labour, unspecified (O66.4) as the "main condition". The code for caesarean section delivery, unspecified (O82.9), may be used as an optional additional code.

Example 24:

Main condition:	Twin pregnancy delivered
Other conditions:	—
Procedure:	Spontaneous delivery

Code twin pregnancy (O30.0) as the "main condition". O84.0 (Multiple delivery, all spontaneous) may be added as an optional additional code.

Example 25:

Main condition:	Term pregnancy delivered of dead fetus, 2800 g
Other conditions:	—
Procedure:	Spontaneous delivery

Code to maternal care for intrauterine death (O36.4) if no specific reason for the fetal death can be determined.

O98-O99 Maternal diseases classifiable elsewhere but complicating pregnancy, childbirth and the puerperium

The subcategories provided should be used as "main condition" codes in preference to categories outside Chapter XV when the conditions being classified have been indicated by the health care practitioner to have complicated the pregnant state, to have been aggravated by the pregnancy, or to have been the reason for obstetric care. The pertinent codes from other chapters may be used as optional additional codes to allow specification of the condition.

Example 26:

Main condition:	Toxoplasmosis
Other conditions:	Pregnancy undelivered
Specialty:	High-risk antenatal clinic

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Code protozoal diseases complicating pregnancy, childbirth and the puerperium (O98.6) as the main condition. B58.9 (Toxoplasmosis, unspecified) may be used as an optional additional code to identify the specific organism.

Chapter XVIII: Symptoms, signs and abnormal clinical and laboratory findings, not elsewhere classified

Categories from this chapter should not be used as "main condition" codes unless the symptom, sign or abnormal finding was clearly the main condition treated or investigated during an episode of care and was unrelated to other conditions recorded by the health care practitioner. See also Rule MB3 (4.4.3) and the introduction to Chapter XVIII in Volume 1 for further information.

Chapter XIX: Injury, poisoning and certain other consequences of external causes

Where multiple injuries are recorded and no one of these has been selected as the "main condition", code to one of the categories provided for statements of multiple injuries of:

- same type to the same body region (usually fourth character .7 in categories S00-S99);
- different types to the same body region (usually fourth character .7 in the last category of each block, i.e. S09, S19, S29, etc.); and
- same type to different body regions (T00-T05).

Note the following exceptions:

- for internal injuries recorded with superficial injuries and/or open wounds only, code to internal injuries as the "main condition";
- for fractures of skull and facial bones with associated intracranial injury, code to the intracranial injury as the "main condition";
- for intracranial haemorrhage recorded with other injuries to the head only, code to intracranial haemorrhage as the "main condition"; and
- for fractures recorded with open wounds of the same location only, code to fracture as the "main condition".

When the multiple injury categories are used, codes for any individual injuries listed may be used as optional additional codes. In the case of the exceptions mentioned, in addition to the main condition code, the associated injury may be identified either by an optional additional code or by one of the digits provided for this purpose.

Example 27: Main condition: Injury of bladder and urethra
Other conditions: —

Code to injury of multiple pelvic organs (S37.7) as the "main condition". S37.2 (Injury of bladder) and S37.3 (Injury of urethra) may be used as optional additional codes.

Example 28: Main condition: Open intracranial wound with cerebellar haemorrhage
Other conditions: —

Code to traumatic cerebellar haemorrhage (S06.8) as the "main condition". The open intracranial wound may be indicated, if desired, by the addition of the code S01.9 (Open wound of head, part unspecified) or by the addition of the digit 1 (with open intracranial wound) to code S06.8

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(S06.8.1).

T90-T98 Sequelae of injuries, of poisoning and of other consequences of external causes

These codes are not to be used as the preferred codes for "main condition" if the nature of the residual conditions is recorded. When coding to the residual condition, T90-T98 may be used as optional additional codes.

Chapter XX: External causes of morbidity and mortality

These codes are not to be used as "main condition" codes. They are intended for use as optional additional codes to identify the external cause of conditions classified in Chapter XIX, and may also be used as optional additional codes with conditions classified in any other chapter but having an external cause.

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